MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Outotec
LARSEN & TOUBRO

Project: Mansourah Massarah Gold Project

Customer: MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title: CALCULATION

3120 FOUNDATION DESIGN OF TRANSFER POINT TP-1

DESIGN CALCULATION

AS BUILT

MD-216-3120-EG-CV-CAL-1002

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
DO1	12-10-2023	AS BUILT	RPK	SBY	PRI	
C01	11-05-2020	ISSUED FOR CONSTRUCTION	RPK	SBY	PRI	
A01	10-04-2020	FIRST ISSUE	RPK	SBY	PRI	

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Ma'aden Document Number:

MD-216-3120-EG-CV-CAL-1002 D01

Outotec Document Number:

900707 GAU01 ZC01 CED01 31201 OU100668051 2

L&T Document Number:

O19045 100 C K 24111 001 2

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Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01
Date	10-04-2020	11-05-2020	12-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
Date	10-04-2020	11-05-2020	12-10-2023
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Prepared By	RPK	RPK	RPK
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Approved	PRI	PRI	PRI

REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
SBC 301 :2007	Saudi Structural - Loading & Forces Code
SBC 304 :2007	Saudi Code for Concrete Structures

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Checked	SBY	SBY	SBY	Checked	SBY	SBY	SBY
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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
RL	:	Reduced Level
C.L.	:	Centre Line
CJ	:	Construction Joint
FDN	:	Foundation
TOC	:	Top of concrete
FGL	:	Finished Ground Level
FFL	:	Finished Floor Level
BOF	:	Bottom of footing
TYP	:	Typical
TOC	:	Top of concrete
IP	:	Insert Plate
RCC	:	Reinforced Cement Concrete
MKD.	:	Marked
THK.	:	Thick

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1. INTENT

The document is prepared for Concrete design calculation of TP-1
TP-1 is the transfer point for conveyor 3110-CNV-001 and conveyor 3110-CNV-002

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.
MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.
The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

This document is prepared for foundation design of TP-1. All the loads applied to the structure for analysis & design followed Civil -Structural Design Criteria and SBC 301:2007
The column reinforcement design has been performed in Staad-Pro Software. The footing and tie beam design have been performed in MS Excel.

4. CONCRETE STRUCTURE

4.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304 and ACI 318M.

4.2 Materials

Concrete Strength of 28 days
· Foundation and Structural Members: 35 MPa
· Grade slab and pavement: 30 MPa
· Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade
· Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
· Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

4.3 Foundations for Structures

Foundation shall be Isolated and combined Shallow Concrete foundation. The design calculations are prepared as per recommendation of standard SBC-304 Clause 13.3.

4.4 Minimum Concrete Cover

Items	Minimum Cover(mm)
Foundation (Side & Bottom)	75
Foundation (Top)	50
Column & Pedestal	50
Beam	50
Slab	40

4.5 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 500 kN/m2 at 1m & 3m depth below FGL (ie. RL 911.500m & RL 909.500m respectively) refer Table 4 - BH-17,18,19,25,29 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

4.6 Reference

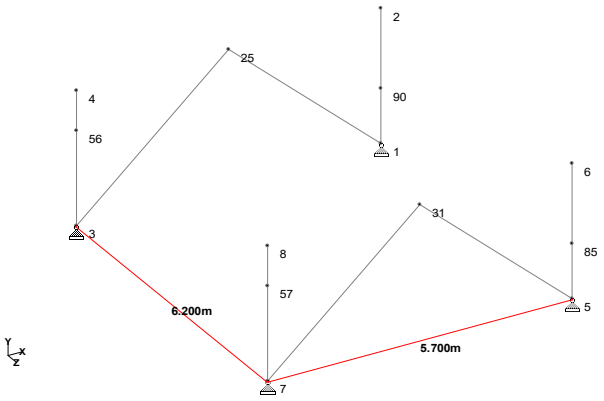
Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

5 INPUT LOADS FROM SUPERSTRUCTURE

For load calculation of superstructure refer Document No: MD-216-3120-EG-ST-CAL-1004

5.1 Support Reaction at Grid 3110/B' - 3110/1"

Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN
3	1 DEAD LOAD	3.1	18.0	3.7
	2 LIVE LOAD	9.9	38.1	7.4
	3 SDL1 - EQUIPMENT DEAD	-8.1	55.1	19.3
	4 SDL2 - EQUIPMENT DEAD	-17.0	21.9	12.6
	5 WL IN +X	-6.5	-6.2	-0.8
	6 WL IN -X	6.5	6.2	0.8
	7 WL IN +Z	0.8	-5.7	-6.4
	8 WL IN -Z	-0.8	5.7	6.4
	9 TEMERATURE LOAD 1	0.5	0.0	1.7
	10 TEMERATURE LOAD 2	-0.5	0.0	-1.7
	11 EARTHQUAKE LOAD (+X)	-1.6	-1.5	0.0
	12 EARTHQUAKE LOAD (+Z)	0.5	-0.4	-0.9



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Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

5.2 Support Reaction at Grid 3110/B' - 3110/1'

Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN
1	1 DEAD LOAD	-3.0	24.8	3.6
	2 LIVE LOAD	-9.8	51.9	4.7
	3 SDL1 - EQUIPMENT DEAD	-25.4	106.1	21.1
	4 SDL2 - EQUIPMENT DEAD	-34.1	114.5	33.1
	5 WL IN +X	-6.5	6.2	0.6
	6 WL IN -X	6.5	-6.2	-0.6
	7 WL IN +Z	0.9	-7.2	-7.1
	8 WL IN -Z	-0.9	7.2	7.1
	9 TEMERATURE LOAD 1	-0.5	0.0	1.7
	10 TEMERATURE LOAD 2	0.5	0.0	-1.7
	11 EARTHQUAKE LOAD (+X)	-1.6	1.5	0.0
	12 EARTHQUAKE LOAD (+Z)	0.5	-2.6	-2.5

5.3 Support Reaction at Grid 3110/D' - 3110/1"

Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN
7	1 DEAD LOAD	2.3	18.5	-3.7
	2 LIVE LOAD	6.9	36.2	-7.4
	3 SDL1 - EQUIPMENT DEAD	-25.6	47.7	-22.7
	4 SDL2 - EQUIPMENT DEAD	-30.9	4.4	-10.2
	5 WL IN +X	-7.5	-7.6	0.8
	6 WL IN -X	7.5	7.6	-0.8
	7 WL IN +Z	-0.8	5.8	-6.0
	8 WL IN -Z	0.8	-5.8	6.0
	9 TEMERATURE LOAD 1	0.5	0.0	-1.7
	10 TEMERATURE LOAD 2	-0.5	0.0	1.7
	11 EARTHQUAKE LOAD (+X)	-1.7	-1.7	0.0
	12 EARTHQUAKE LOAD (+Z)	-0.5	0.4	-0.9

5.4 Support Reaction at Grid 3110/D' - 3110/1'

Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN
5	1 DEAD LOAD	-2.3	26.2	-3.6
	2 LIVE LOAD	-6.9	51.1	-4.7
	3 SDL1 - EQUIPMENT DEAD	-25.9	139.0	-39.7
	4 SDL2 - EQUIPMENT DEAD	-31.1	110.2	-17.5
	5 WL IN +X	-7.5	7.6	-0.6
	6 WL IN -X	7.5	-7.6	0.6
	7 WL IN +Z	-0.9	7.1	-6.7
	8 WL IN -Z	0.9	-7.1	6.7
	9 TEMERATURE LOAD 1	-0.5	0.0	-1.7
	10 TEMERATURE LOAD 2	0.5	0.0	1.7
	11 EARTHQUAKE LOAD (+X)	-1.7	1.7	0.0
	12 EARTHQUAKE LOAD (+Z)	-0.5	2.6	-2.5

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5.5 LOAD COMBINATIONS

5.5.1 Load Combination for Base pressure calculation

(As per Section 2.4 of SBC 301 -2007)

Load Cases	Reference
LOAD 301 (DL + SDL1)	(Eq. 2.4.1-1 of SBC 301)
LOAD 302 (DL + SDL2)	
LOAD 303 (DL + SDL1 + TL1) + LL	(Eq. 2.4.1-2 of SBC 301)
LOAD 304 (DL + SDL2 + TL1) + LL	
LOAD 305 (DL + SDL1 + TL2) + LL	
LOAD 306 (DL + SDL2 + TL2) + LL	
LOAD 307 (DL+ SDL1) + 0.75 LL + 0.75 WL(+X)	(Eq. 2.4.1-6 of SBC 301)
LOAD 308 (DL+ SDL1) + 0.75 LL + 0.75 WL(-X)	
LOAD 309 (DL+ SDL1) + 0.75 LL + 0.75 WL(+Z)	
LOAD 310 (DL+ SDL1) + 0.75 LL + 0.75 WL(-Z)	
LOAD 311 (DL+ SDL2) + 0.75 LL + 0.75 WL(+X)	
LOAD 312 (DL+ SDL2) + 0.75 LL + 0.75 WL(-X)	
LOAD 313 (DL+ SDL2) + 0.75 LL + 0.75 WL(+Z)	
LOAD 314 (DL+ SDL2) + 0.75 LL + 0.75 WL(-Z)	
LOAD 315 (DL+ SDL1) + 0.75 LL + 0.525 EL(+X)	(Eq. 2.4.1-6 of SBC 301)
LOAD 316 (DL+ SDL1) + 0.75 LL + 0.525 EL(-X)	
LOAD 317 (DL+ SDL1) + 0.75 LL + 0.525 EL(+Z)	
LOAD 318 (DL+ SDL1) + 0.75 LL + 0.525 EL(-Z)	
LOAD 319 (DL+ SDL2) + 0.75 LL + 0.525 EL(+X)	
LOAD 320 (DL+ SDL2) + 0.75 LL + 0.525 EL(-X)	
LOAD 321 (DL+ SDL2) + 0.75 LL + 0.525 EL(+Z)	
LOAD 322 (DL+ SDL2) + 0.75 LL + 0.525 EL(-Z)	
LOAD 323 0.6 (DL+SDL1) + 1.0 WL(+X)	(Eq. 2.4.1-7 of SBC 301)
LOAD 324 0.6 (DL+SDL1) + 1.0 WL(-X)	
LOAD 325 0.6 (DL+SDL1) + 1.0 WL(+Z)	
LOAD 326 0.6 (DL+SDL1) + 1.0 WL(-Z)	
LOAD 327 0.6 (DL+SDL2) + 1.0 WL(+X)	
LOAD 328 0.6 (DL+SDL2) + 1.0 WL(-X)	
LOAD 329 0.6 (DL+SDL2) + 1.0 WL(+Z)	
LOAD 330 0.6 (DL+SDL2) + 1.0 WL(-Z)	
LOAD 331 0.6 (DL+SDL1) + 0.7 EL(+X)	(Eq. 2.4.1-8 of SBC 301)
LOAD 332 0.6 (DL+SDL1) + 0.7 EL(-X)	
LOAD 333 0.6 (DL+SDL1) + 0.7 EL(+Z)	
LOAD 334 0.6 (DL+SDL1) + 0.7 EL(-Z)	
LOAD 335 0.6 (DL+SDL2) + 0.7 EL(+X)	
LOAD 336 0.6 (DL+SDL2) + 0.7 EL(-X)	
LOAD 337 0.6 (DL+SDL2) + 0.7 EL(+Z)	
LOAD 338 0.6 (DL+SDL2) + 0.7 EL(-Z)	

5.5.2 Load Combination for Concrete design

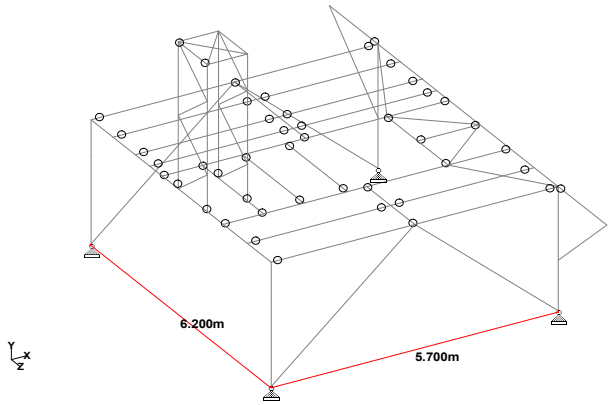
Load Case	Reference
LOAD 401 1.4 (DL + SDL1)	(Eq. 2.3.2-1 of SBC 301)
LOAD 402 1.4 (DL + SDL2)	
LOAD 403 1.4 (DL + SDL1 + TL1) + 1.7 LL	(Eq. 2.3.2-2 of SBC 301)
LOAD 404 1.4 (DL + SDL2 + TL1) + 1.7 LL	
LOAD 405 1.4 (DL + SDL1 + TL2) + 1.7 LL	
LOAD 406 1.4 (DL + SDL2 + TL2) + 1.7 LL	
LOAD 407 1.2 (DL+ SDL1) + 0.5 LL + 1.6 WL(+X)	(Eq. 2.3.2-4 of SBC 301)
LOAD 408 1.2 (DL+ SDL1) + 0.5 LL + 1.6 WL(-X)	
LOAD 409 1.2 (DL+ SDL1) + 0.5 LL + 1.6 WL(+Z)	
LOAD 410 1.2 (DL+ SDL1)+ 0.5 LL + 1.6 WL(-Z)	
LOAD 411 1.2 (DL+ SDL2) + 0.5 LL + 1.6 WL(+X)	
LOAD 412 1.2 (DL+ SDL2) + 0.5 LL + 1.6 WL(-X)	
LOAD 413 1.2 (DL+ SDL2) + 0.5 LL + 1.6 WL(+Z)	
LOAD 414 1.2 (DL+ SDL2)+ 0.5 LL + 1.6 WL(-Z)	
LOAD 415 1.2 (DL+ SDL1) + 0.5 LL + 1.0 EL(+X)	(Eq. 2.3.2-5 of SBC 301)
LOAD 416 1.2 (DL+ SDL1) + 0.5 LL + 1.0 EL(-X)	
LOAD 417 1.2 (DL+ SDL1) + 0.5 LL + 1.0 EL(+Z)	
LOAD 418 1.2 (DL+ SDL1)+ 0.5 LL + 1.0 EL(-Z)	
LOAD 419 1.2 (DL+ SDL2) + 0.5 LL + 1.0 EL(+X)	
LOAD 420 1.2 (DL+ SDL2) + 0.5 LL + 1.0 EL(-X)	
LOAD 421 1.2 (DL+ SDL2) + 0.5 LL + 1.0 EL(+Z)	
LOAD 422 1.2 (DL+ SDL2) + 0.5 LL + 1.0 EL(-Z)	
LOAD 423 0.9 (DL+ SDL1) + 1.6 WL(+X)	(Eq. 2.3.2-6 of SBC 301)
LOAD 424 0.9 (DL+ SDL1) + 1.6 WL(-X)	
LOAD 425 0.9 (DL+ SDL1) + 1.6 WL(+Z)	
LOAD 426 0.9 (DL+ SDL1) + 1.6 WL(-Z)	
LOAD 427 0.9 (DL+ SDL2) + 1.6 WL(+X)	
LOAD 428 0.9 (DL+ SDL2) + 1.6 WL(-X)	
LOAD 429 0.9 (DL+ SDL2) + 1.6 WL(+Z)	
LOAD 430 0.9 (DL+ SDL2) + 1.6 WL(-Z)	
LOAD 431 0.9 (DL+ SDL1) + 1.0 EL(+X)	(Eq. 2.3.2-7 of SBC 301)
LOAD 432 0.9 (DL+ SDL1) + 1.0 EL(-X)	
LOAD 433 0.9 (DL+ SDL1) + 1.0 EL(+Z)	
LOAD 434 0.9 (DL+ SDL1) + 1.0 EL(-Z)	
LOAD 435 0.9 (DL+ SDL2) + 1.0 EL(+X)	
LOAD 436 0.9 (DL+ SDL2) + 1.0 EL(-X)	
LOAD 437 0.9 (DL+ SDL2) + 1.0 EL(+Z)	
LOAD 438 0.9 (DL+ SDL2) + 1.0 EL(-Z)	

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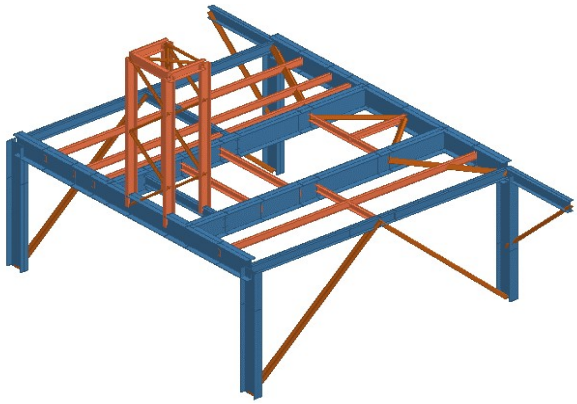
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6.2 Geometry of TP-1 Superstructure

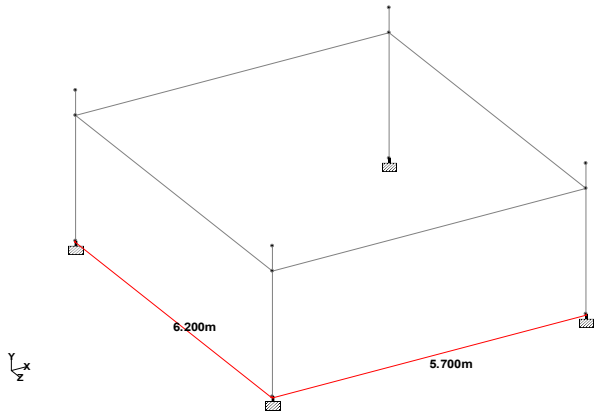


Staad Model of TP-1 Superstructure

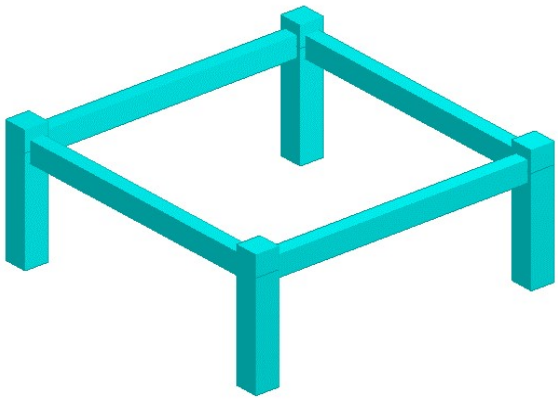


Isometric view of TP-1 Superstructure

6.3 Geometry of TP-1 Foundation



Staad Model of TP-1 Foundation framework



Isometric view of TP-1 Foundation framework

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Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

7. DESIGN OF CONCRETE COMPONENTS

All footings & tie beams have been designed using worksheet in accordance to SBC-304.

All columns have been designed in Staad-Pro by ACI 318M.

For Design Results, Refer Attachment 3 - Staad Output file.

7.1 Design summary of pedestals

7.1.1 Pedestal Reinforcement Summary

Column mark	Dimension (mm)	Reinf. Area Required (mm^2)	Reinf. Area Provided (mm^2)	Reinforcement	Links	Remarks	Numbers
P1	600 X 600	3600	3768	12 No.- D20	3 Nos D10 @ 200c/c	Grid 3110/B'-Grid 3110/1"	1
P1	600 X 600	3600	3768	12 No.- D20	3 Nos D10 @ 200c/c	Grid 3110/B'-Grid 3110/1'	1
P1	600 X 600	3600	3768	12 No.- D20	3 Nos D10 @ 200c/c	Grid 3110/D'-Grid 3110/1"	1
P1	600 X 600	3600	3768	12 No.- D20	3 Nos D10 @ 200c/c	Grid 3110/D'-Grid 3110/1'	1

7.2 Design of Footings

7.2.1 Footing marking, Maximum Base pressure and Reinforcement Summary

Structure Name	Foundation mark	Length (mm)	Width (mm)	Thickness (mm)	Max pressure (kN/m^2)	Top reinforcement	Bottom reinforcement	F.O.S.	Numbers
Column at 3110/B' - 3110/1"	F1	2200	2200	400	92	D12 @ 150c/c B/W	D12 @ 150c/c B/W	0.18	1
Column at 3110/B' - 3110/1'	F1	2200	2200	400	129	D12 @ 150c/c B/W	D12 @ 150c/c B/W	0.25	1
Column at 3110/D' - 3110/1"	F1	2200	2200	400	90	D12 @ 150c/c B/W	D12 @ 150c/c B/W	0.17	1
Column at 3110/D' - 3110/1'	**F1**	**2200**	**2200**	**400**	**137**	**D12 @ 150c/c B/W**	**D12 @ 150c/c B/W**	**0.26**	**1**

The design of the most critical foundation of the group is attached in the next pages.

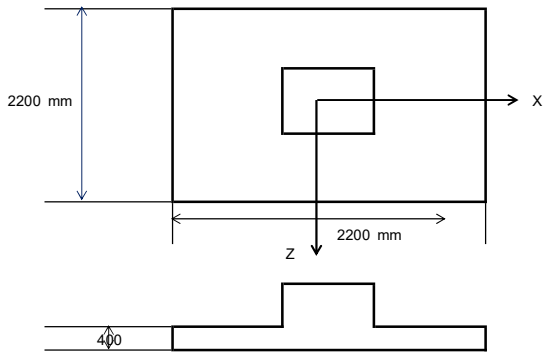
Global Project ID	900707
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Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T	0	1	2
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Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

7.2.2 Design of footing F1 at Grid 3110/D'-3110/1'

*inputs

Finished Ground level	F.G.L	=	0.000	m	Weight of soil	=	80.64 KN
Bottom of foundation	B.O.F	=	-1.400	m	Weight of foundation	=	46.46 KN
Top of footing (staad support)	T.O.F	=	-1.000	m	Uplift	=	0.00 KN
Ground Water Level	G.W.L	=	-10.000	m	Surcharge	=	0 KN
Surcharge	Ws	=	0.000	KN/m^2	Pressure on top for tension	=	26.3 KN/m^2
Length of footing	Lz	=	2.20	m			
Width of footing	Lx	=	2.20	m			
Thickness of footing	D	=	0.40	m			
Depth of pedestal	Bx	=	0.60	m			
Width of pedestal	Bz	=	0.60	m			
Height of Load point from bottom of footing	H	=	0.400	m			
Section Modulus of Pedestal	Zxx	=	1.77	m^3			
Section Modulus of Pedestal	Zzz	=	1.77	m^3			
Allowable Net Bearing Capacity		=	500	KN/m^2			
Gross Bearing Capacity		=	525.2	KN/m^2			
Grade of concrete		=	35	N/mm^2			
Grade of steel		=	420	N/mm^2			
Clear cover		=	75	mm			
Top Reinfocement		=	12	mm			
Bottom Reinforcement		=	12	mm			
Co-efficient of friction between soil & concrete		=	0.3				
Unit wt of soil		=	18	KN/m^3			
Unit wt of concrete		=	24	KN/m^3			



$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$

β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301	31.1	212.64	8.8	24.94	-0.1	-63.3	339.7	119.90	20.49	1	0.23
302	36.5	182.16	-7.0	-13.49	0.24	-75.5	309.3	114.02	13.78	1	0.22
303	29.4	263.73	7.3	24.52	-0.06	-62.7	390.8	129.89	31.61	1	0.25
304	34.7	233.25	-8.4	-13.89	0.24	-74.9	360.4	124.47	24.44	1	0.24
305	29.6	263.73	8.1	24.74	-0.06	-62.8	390.8	130.06	31.44	1	0.25
306	35.0	233.25	-7.7	-13.68	0.2	-75.0	360.4	124.40	24.51	1	0.24
307	35.5	257.67	7.9	24.71	-0.05	-75.6	384.8	136.04	22.96	1	0.26
308	24.3	244.24	8.1	24.71	-0.07	-50.1	371.3	118.88	34.57	1	0.23
309	30.5	257.2	13.2	36.61	-0.05	-64.3	384.3	136.24	22.56	1	0.26
310	29.3	244.71	2.8	12.81	-0.08	-61.5	371.8	118.68	34.97	1	0.23
311	40.9	227.2	-7.9	-13.71	0.3	-87.8	354.3	130.41	16.00	1	0.25
312	29.7	213.77	-7.7	-13.72	0.23	-62.3	340.9	113.26	27.60	1	0.22
313	35.9	226.73	-2.6	-1.81	0.26	-76.4	353.8	117.20	29.01	1	0.22
314	34.6	214.23	-13.0	-25.61	0.2	-73.7	341.3	126.46	14.59	1	0.24
315	30.8	252	8.0	24.71	-0.06	-64.9	379.1	128.84	27.81	1	0.25
316	29.0	249.91	8.0	24.70	-0.06	-60.8	377.0	126.07	29.72	1	0.24
317	30.2	252.56	9.2	27.65	-0.07	-63.4	379.7	129.77	27.12	1	0.25
318	29.7	249.34	6.7	21.77	-0.05	-62.3	376.4	125.14	30.41	1	0.24
319	36.2	221.52	-7.8	-13.71	0.3	-77.1	348.6	123.21	20.85	1	0.23
320	34.3	219.44	-7.8	-13.72	0.24	-73.0	346.5	120.45	22.75	1	0.23
321	35.5	222.09	-6.5	-10.78	0.23	-75.6	349.2	120.83	23.46	1	0.23
322	35.0	218.87	-9.1	-16.65	0.26	-74.5	346.0	122.82	20.14	1	0.23
323	26.2	136.53	5.1	14.96	-0.02	-55.0	263.6	93.88	15.06	1	0.18
324	11.2	118.63	5.4	14.96	-0.1	-20.9	245.7	71.00	30.54	1	0.14
325	19.5	135.91	12.2	30.83	-0.02	-39.8	263.0	94.16	14.52	1	0.18
326	17.9	119.25	-1.7	-0.91	-0.06	-36.1	246.4	71.75	30.05	1	0.14
327	29.4	118.25	-4.3	-8.09	0.16	-62.3	245.4	90.35	11.03	1	0.17
328	14.4	100.35	-4.1	-8.08	0.1	-28.3	227.5	67.47	26.52	1	0.13
329	22.7	117.63	2.7	7.79	0.17	-47.1	244.7	81.52	19.61	1	0.16
330	21.1	100.97	-11.1	-23.96	0.12	-43.4	228.1	85.08	9.17	1	0.16
331	19.9	128.97	5.3	14.96	-0.04	-40.7	256.1	84.28	21.53	1	0.16
332	17.5	126.19	5.3	14.95	-0.04	-35.2	253.3	80.60	24.07	1	0.15
333	19.0	129.73	7.0	18.88	-0.1	-38.7	256.8	85.53	20.60	1	0.16
334	18.3	125.43	3.6	11.04	-0.02	-37.2	252.5	79.36	25.00	1	0.15
335	23.1	110.69	-4.2	-8.08	0.15	-48.0	237.8	80.75	17.51	1	0.15
336	20.7	107.91	-4.2	-8.10	0.14	-42.5	235.0	77.08	20.03	1	0.15
337	22.2	111.45	-2.5	-4.17	0.13	-46.0	238.6	77.58	21.00	1	0.15
338	21.5	107.15	-5.9	-12.01	0.2	-44.5	234.3	80.24	16.56	1	0.15

Max P_{max}	=	136.24	KN/m^2
Max P_{min}	=	9.17	KN/m^2
Resultant F.O.S	=	0.25	

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BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401	43.6	297.7	12.3	34.91	-0.1	-88.58	131.09	-8.08
402	51.1	255.0	-9.8	-18.88	0.34	-105.63	122.85	-17.47
403	40.7	384.5	9.9	34.24	-0.08	-87.61	148.11	10.79
404	48.1	341.9	-12.1	-19.54	0.34	-104.67	140.63	0.65
405	41.0	384.5	11.0	34.55	-0.08	-87.72	148.34	10.56
406	48.5	341.9	-11.0	-19.25	0.34	-104.77	140.52	0.75
407	48.5	295.0	9.8	29.77	-0.05	-102.88	135.70	-13.79
408	24.6	266.4	10.2	29.77	-0.1	-48.43	99.10	10.98
409	37.9	294.0	21.1	55.17	-0.04	-78.64	136.15	-14.65
410	35.2	267.4	-1.1	4.38	-0.11	-72.67	98.66	11.83
411	54.9	258.5	-9.1	-16.34	0.32	-117.50	128.82	-22.01
412	31.0	229.8	-8.7	-16.33	0.27	-63.05	92.21	2.75
413	44.3	257.5	2.2	9.06	0.33	-93.26	110.85	-4.46
414	41.6	230.8	-20.0	-41.73	0.26	-87.29	120.39	-25.01
415	38.3	282.7	10.0	29.78	-0.07	-79.59	120.04	-3.22
416	34.8	278.7	10.0	29.77	-0.08	-71.72	114.77	0.40
417	37.0	283.8	12.4	35.37	-0.1	-76.76	121.81	-4.55
418	36.1	277.6	7.6	24.18	-0.05	-74.55	113.00	1.73
419	44.7	246.1	-8.9	-16.32	0.29	-94.21	113.14	-11.43
420	41.2	242.2	-8.9	-16.34	0.29	-86.34	107.89	-7.82
421	43.4	247.2	-6.5	-10.73	0.27	-91.37	108.61	-6.46
422	42.5	241.1	-11.3	-21.93	0.31	-89.17	112.41	-12.80
423	40.0	205.7	7.7	22.44	-0.03	-84.17	102.57	-17.57
424	16.1	177.1	8.1	22.44	-0.08	-29.72	65.97	7.19
425	29.4	204.7	19.0	47.83	-0.02	-59.93	103.02	-18.43
426	26.7	178.0	-3.2	-2.95	-0.09	-53.96	68.85	4.72
427	44.8	178.3	-6.5	-12.13	0.24	-95.14	97.28	-23.61
428	20.9	149.6	-6.1	-12.13	0.19	-40.67	60.67	1.16
429	34.1	177.3	4.8	13.26	0.25	-70.90	84.05	-10.79
430	31.5	150.6	-17.3	-37.53	0.18	-64.92	88.84	-26.60
431	29.8	193.4	7.9	22.46	-0.05	-60.88	86.91	-7.01
432	26.3	189.4	7.9	22.44	-0.06	-53.01	81.64	-3.38
433	28.5	194.4	10.3	28.04	-0.08	-58.04	88.68	-8.33
434	27.5	188.3	5.5	16.85	-0.04	-55.84	79.87	-2.06
435	34.6	165.9	-6.3	-12.13	0.22	-71.85	81.60	-13.04
436	31.1	162.0	-6.3	-12.15	0.22	-63.97	76.35	-9.43
437	33.3	167.0	-3.8	-6.54	0.2	-69.00	77.08	-8.06
438	32.3	160.9	-8.7	-17.73	0.24	-66.81	80.88	-14.40

Max Net factored B.P = 148.34 KN/m^2

DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 0.8 m

Moment about Z: Muz = 47.47 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 694.79 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 396.7 mm^2

Effective Depth Along X = 319 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

or $A_{stmin} = 0.0014bd$

A_{stmin} = 574.2 mm^2

A_{req} = 396.7 mm^2

Required Spacing = **197.0** mm

Length of cantilever along Z = 0.8 m

Moment about X: Mux = 47.47 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 643.50 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 412.5 mm^2

Effective Depth Along X = 307 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

or $A_{stmin} = 0.0014bd$

A_{stmin} = 552.6 mm^2

A_{req} = 412.5 mm^2

Required Spacing = **204.7** mm

Provide Bottom Reinforcement = **12 mm dia @ 150 c/c**

Ast prov = 754.0 mm^2

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Top Reinforcement

Length of cantilever along X = 0.8 m
Moment about Z: Muz = 0.00 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 694.79 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 0.0 mm^2

Effective Depth Along X = 319 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of
A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% o
or
A_{stmin} = 50% of 0.0014bd more prov i bottom

Table 24.4.3.2 SBC 304

A_{stmin} = 287.1 mm^2
A_{req} = 0.0 mm^2

Required Spacing = **393.9** mm

Length of cantilever along Z = 0.8 m
Moment about X: Mux = 0.00 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 643.50 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 0.0 mm^2

Effective Depth Along Z = 307 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of
A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% o
or
A_{stmin} = 50% of 0.0014bd more prov i bottom

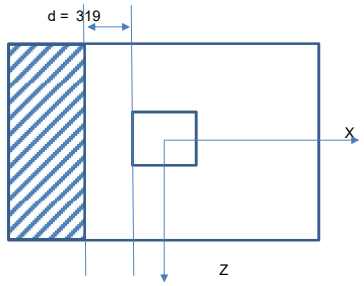
Table 24.4.3.2 SBC 304

A_{stmin} = 276.3 mm^2
A_{req} = 0.0 mm^2

Required Spacing = **409.3** mm

Provide top Reinforcement = **12 mm dia @ 150 c/c** Ast prov = 754.0 mm^2

ONE WAY SHEAR CKECK



Along X Direction Cantilever: L = 0.481 m
Reinforcement ratio: ρ_w = 0.0024
Effective Depth Along X: d = 319 mm
Shear Force at d distance: V_u = 71.35 KN
Allowable shear strength: φV_c = 236.08 KN O.K

Least of

i) $V_c = \left(0.16\lambda\sqrt{f'_c} b_w d + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ *Table 22.5.5.1 SBC 304*

λ = 1 Normal concrete
Mu = 17.16 KNm
φ = 0.75 for shear
φV_c = 239.22 KN

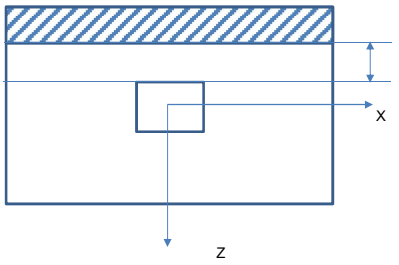
ii) $V_c = \left(0.16\lambda\sqrt{f'_c} b_w d + 17\rho_w\right) b_w d$
φV_c = 236.080808 KN *Table 22.5.5.1 SBC 304*

iii) $V_c = 0.29\lambda\sqrt{f'_c} b_w d$
= 410.472 KN

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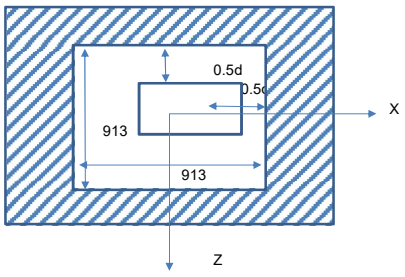
ONE WAY SHEAR CKECK



Along Z Direction	Cantilever	L	=	0.493	m	Least of	i)	V_c	= $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$
Reinforcement ratio		ρ_w	=	0.0025					Table 22.5.5.1 SBC 304
Effective Depth Along Z		d	=	307	mm			λ	= 1 Normal concrete
Shear Force at d distance		V_u	=	73.13	KN			Mu	= 18.03 KNm
Allowable shear strength		φVc	=	227.56	KN	O.K		φ	= 0.75 for shear
								φVc	= 229.92 KN
							ii)	V_c	= $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$
								φVc	= 227.561653 KN
							iii)	V_c	= $0.29\lambda\sqrt{f_c'}b_w d$
								φVc	= 395.03 KN

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area		=	4.84	m2
Punching area		=	0.83	m2
	λ	=	1	
	β	=	1	
	αs	=	40	
	b0	=	3652.0	mm

Allowable shear stress — least of — Table 22.6.5.2 SBC 304

v_c	$0.33\lambda\sqrt{f_c'}$	=	1.95 N/mm²
v_c	$0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$	=	3.02 N/mm²
v_c	$0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$	=	2.67 N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

Perimeter at 0.5d from column face	u	=	3.65	m	Vc	=	$v_c b_0 d$
Depth at section	d	=	313	mm		=	2231.63 KN
Shear Force at section	Vu	=	594.3	KN	φ	=	0.75 for shear
Allowable shear strength	φVc	=	1673.73	KN O.K	φV_n	=	φVc
					φVc	=	1673.7 KN

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

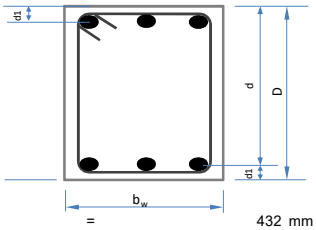
Ma'aden Revision No:	A01	C01	D01
Date	10-04-2020	11-05-2020	12-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
Date	10-04-2020	11-05-2020	12-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

7.3 Design of Tie Beams

7.3.1 Beam marked TB1

Staad beam number - 123,125,127,128



Width of beam	=	**300**	mm	
Depth of beam	=	**500**	mm	
Grade of steel	=	420	MPa	
Grade of concrete	=	35	MPa	
Cover	=	50	mm	
Stress reduction factor for flexure	=	0.9		*Table 21.2.2 SBC 304*
Stress reduction factor for shear	=	0.75		*Table 21.2.1 SBC 304*
Stress reduction factor for torsion	=	0.75		*Table 21.2.1 SBC 304*
Effective Depth	=	432	mm	
Dia of longitudinal bar	=	**16**	mm	
Dia of stirrup bar	=	**10**	mm	

d	=	432	mm
d'	=	68	mm
h	=	120	mm

ANALYSIS RESULTS

Analysis results as obtained from staad

B.M at support	Mz	L/C -	411 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+X)	=	**36** KNm
B.M at span	Mz	L/C -	411 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+X)	=	**14** KNm
Shear Force	Fy	L/C -	402 1.4 (DL+SDL2)	=	**22** KN

DESIGN FOR FLEXURE

Support Moment

Mu lim = $\emptyset\, 0.85\, f_c'\beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 432 x 432 (1 - 0.8/2 x 0.375)

= 382.3 KNm

Singly reinforced beam

$\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

b_1 = 0.80 *Table 22.2.2.4.3 SBC 304*

Mu = 36.37 KNm

Mu_2 = Mu - Mu_{lim} = 36.37 - 382.3

= 0.00 KNm

$A_{st1} = \frac{0.85\, f_c' bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c' bd^2}}\right)$

= 0.85 x 35x 300 x 432 x (1-√(1-(2.35 x 36.4 x 10^6)/(0.9 x 35 x 300 x 432^2)))/420

= 225.2 mm^2

$A_{st2} = \frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d - d')}$

= 0 x 10^6/(0.9 x 420 x (432 - 68))

= 0 mm^2

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

e_{sc} = 0.0017

Asc = 0.0000 mm^2

= 456.4 mm^2 or 432 mm^2 = 456.4 mm^2 *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 456.4 + 0.00 + 0

= 456.4 mm^2

Top layer1	**3**	*nos*	**16**	*mm*	*dia*	+		*nos*		*mm* *dia*
Top layer2		*nos*		*mm*	*dia*	+		*nos*		*mm* *dia*

Ast provided = 603.1857895 mm^2 Safe

ρ_w = 0.00465421

Global Project ID	900707
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Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	10-04-2020	11-05-2020	12-10-2023				Date	10-04-2020	11-05-2020	12-10-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	RPK	RPK	RPK				Prepared By	RPK	RPK	RPK			
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

Span Moment

Mu lim = $\emptyset\, 0.85\, f_c'\, \beta_1 \left(\frac{c}{d}\right)_{max} bd^2 \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right)$

For tension-controlled section & following the assuptions in clause 22.2 of SBC 304

= 0.9 x 0.85 x 35x 0.8 x 0.375 x 300 x 432 x 432 (1 - 0.8/2 x 0.375)

= 382.3 KNm

$\frac{c}{d}$ max = $\frac{0.003}{0.003 + 0.005}$ = 0.375

Singly reinforced beam

b_1 = 0.80

Mu = 14.28 KNm

Mu_2 = Mu - Mu_{lim} = 14.28 - 382.3

= 0 KNm

A_{st} = $\frac{0.85\, f_c'\, bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35\, M_u}{\emptyset\, f_c'\, bd^2}}\right)$

A_{st2} = $\frac{M_u - M_{u\,lim}}{\emptyset\, f_s'(d - d')}$

= 0.85 x 35x 300 x 432 x (1-√(1-(2.35 x 14.3 x 10^6)/(0.9 x 35 x 300 x 432^2)))/420

= 87.8 mm²

= 0 x 10^6/(0.9 x 420 x (432 - 68))

= 0 mm²

e_{sc} = 0.0017

Min reinforcement = max of $\frac{0.25\sqrt{f_c'}}{f_y} b_w d$ or $\frac{1.4}{f_y} b_w d$

Asc = 0.0000 mm²

= 456.4 mm² or 432 mm² = 456.383298 mm² *clause 9.6.1.2 of SBC 304*

Total Ast Req = A_{st1} + A_{st2} + A_{st3}

= 456.4 + 0 + 0

= 456.4 mm²

bottom layer1	3	nos	16	mm	dia	+		nos		mm dia
bottom layer2		nos		mm	dia	+		nos		mm dia

Ast provided = 603.1857895 mm² Safe

ρ_w = 0.004654211

DESIGN FOR SHEAR

Allowable shear force = Minimum of the following Vc

Corresponding to table 22.5.5.1 of SBC 304

$V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ = 125.3 KN

λ = 1 Normal concrete

$V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$ = 132.9 KN

$V_c = 0.29\lambda\sqrt{f_c'}\, b_w d$ = 222.3 KN

fV_c = 0.75 x 125.4 = 94.01 KN > 21.94 KN *Shear Reinforcement not required*

Vs = $V_U - \emptyset V_C = \frac{\emptyset\, A_v f_y d}{s}$

$\frac{A_v}{S} = \frac{V_U - \emptyset V_C}{\emptyset\, f_y d}$ = ((21.94-94.1) x 10^3)/(0.75 x 420 x 432) = 0 mm²/mm

No. of legs = 2 Spacing provided = 200 mm

Provide 2L -10 mm stirrups @ 200 C/C

Spacing of stirrup is minimum of the following

i) Shear Reinforcement		s =	0 mm	*if Vu > φVc*
ii) Min Asv	= $0.062\sqrt{f_c'}\frac{b_w s}{f_t}$	s =	599.5455737 mm	*Corresponding to table 9.6.3.3 of SBC 304*
or Min Asv	= $0.65\frac{b_w s}{f_{yt}}$	s =	338.3253627 mm	*Corresponding to table 9.6.3.3 of SBC 304*
iii) effective depth/2		s =	216 mm	*Corresponding to table 9.7.6.2.2 of SBC 304*
iv) Combined shear & torsion		s =	0 mm	

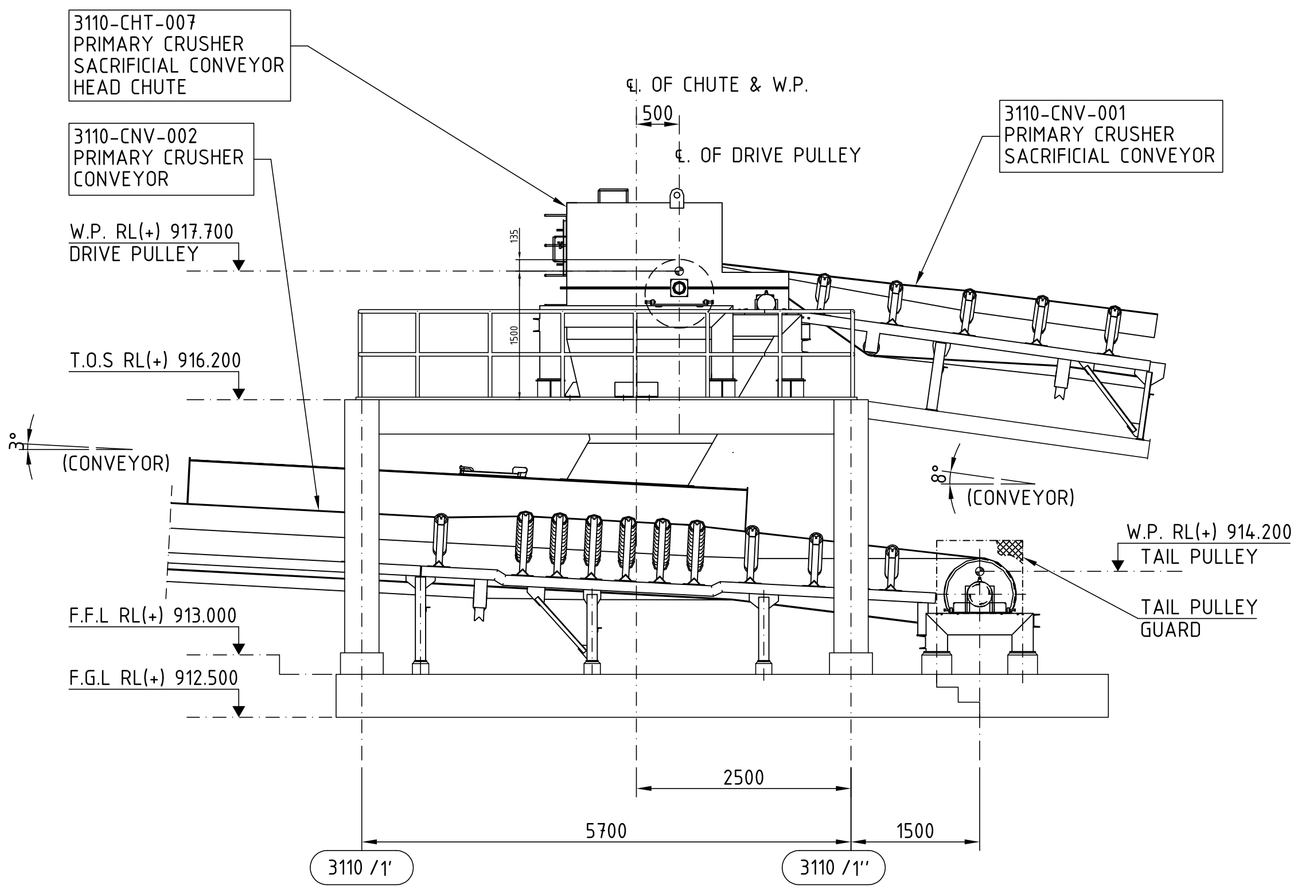
Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
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Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
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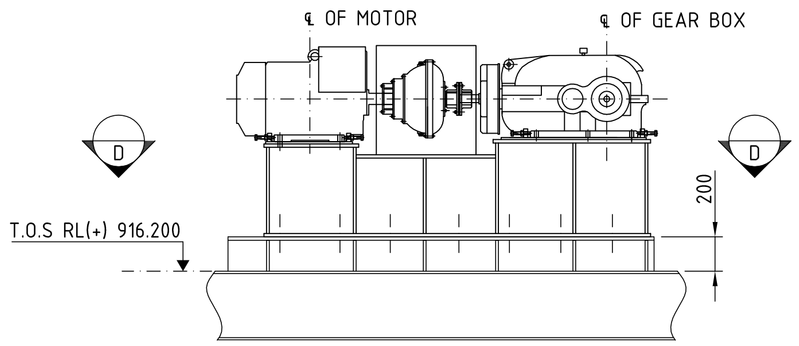
ATTACHMENT 1 **MECHANICAL GENERAL ARRANGEMENT DRAWINGS**

DRAWING NUMBER MD-216-3110-EG-ME-ELV-1011 REV D01



ELEVATION A-A

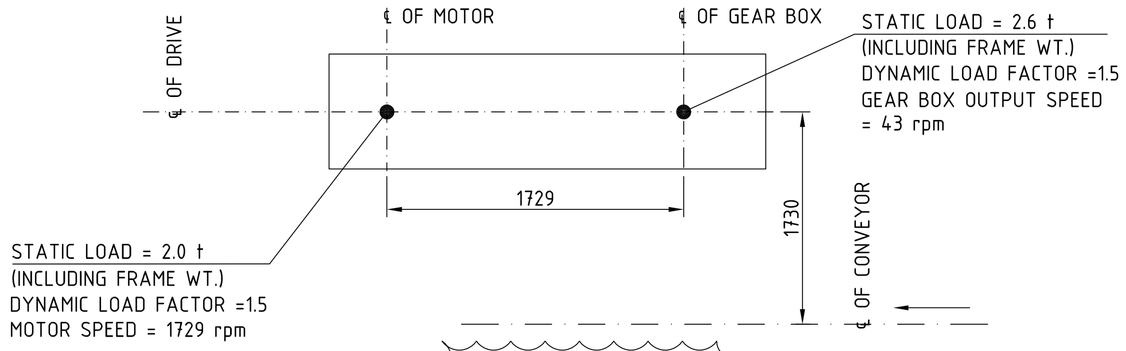
SCALE 1:50



VIEW C-C

SCALE 1:25

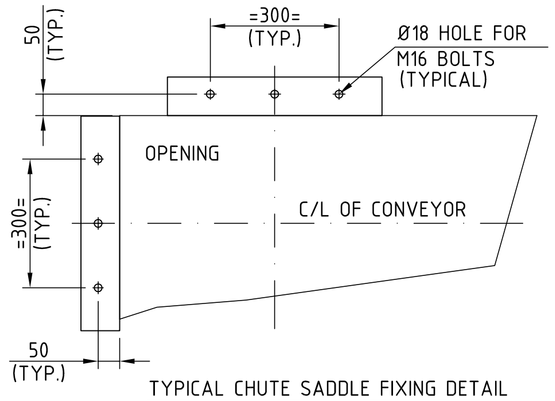
3110-501 HOLD



VIEW D-D

SCALE 1:25

3110-501 HOLD



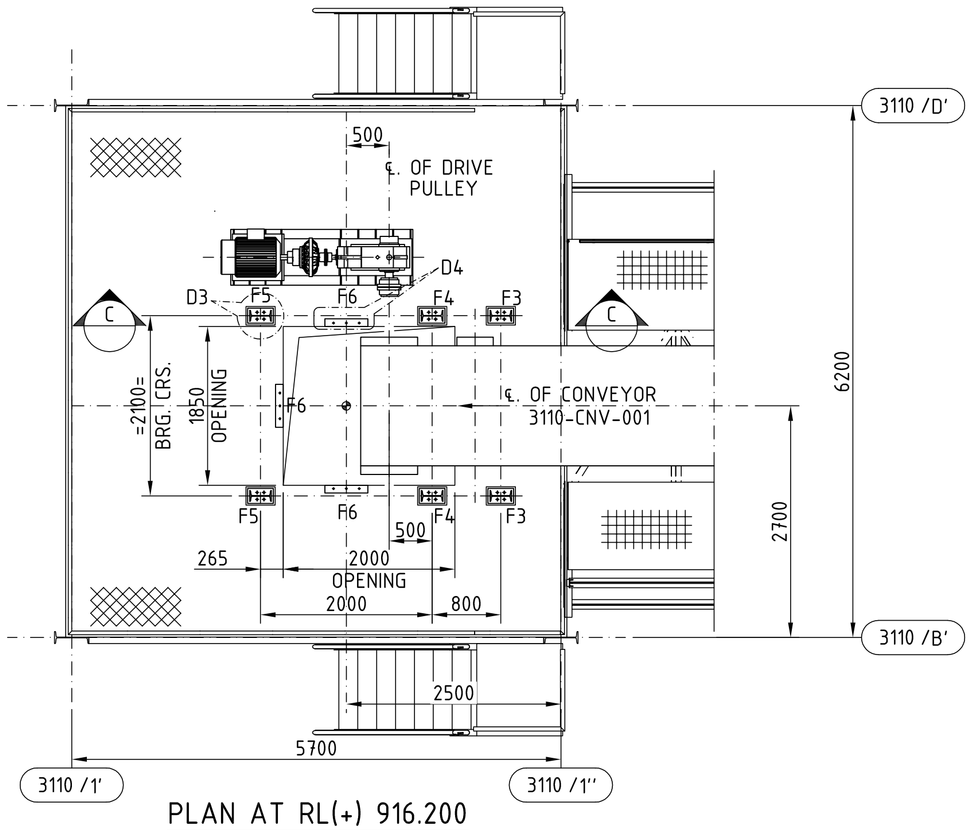
TYPICAL CHUTE SADDLE FIXING DETAIL

NOTE:

1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+)913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL,BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN TENTATIVE.

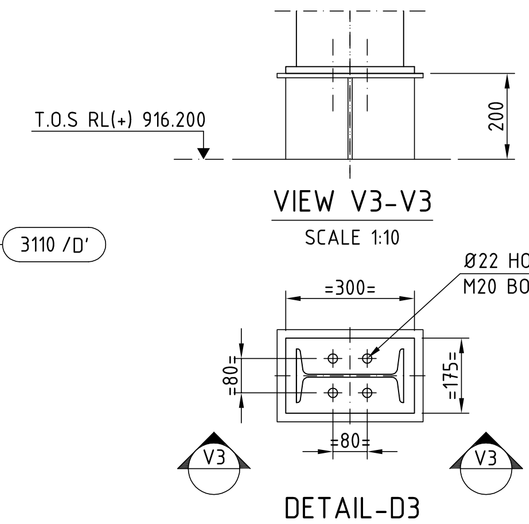
ABBREVIATION	
CONV.	CONVEYOR
W.P.	WORKING POINT
℄	CENTER LINE
CRS.	CENTERS
B/W	BELT WIDTH
rpm	REVOLUTIONS PER MINUTE
BRG.	BEARING
SQ.	SQUARE
WT.	WEIGHT
PROJ.	PROJECTION
TP	TRANSFER POINT

CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL



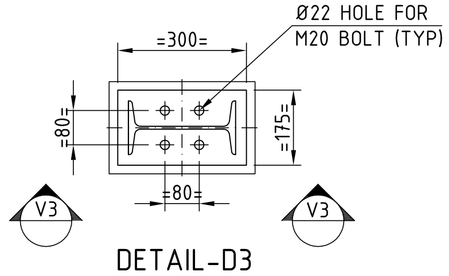
PLAN AT RL(+) 916.200

SCALE 1:50



VIEW V3-V3

SCALE 1:10



DETAIL-D3

SCALE 1:10

TABLE-4 : FOR CHUTE SADDLE LOAD

POINTS	ELEVATIONS IN "m"	V (IN "t")
F6	(+)916.200	(+)1.5

SIGN CONVENTION

(V) VERTICAL LOAD : UPWARD (-)
: DOWNWARD (+)

(H) HORIZONTAL LOAD : TOWARDS HEAD PULLEY=(+)
: TOWARDS TAIL PULLEY=(-)
: REVERSIBLE LOAD=(±)

Note: Wind load & seismic to be considered during civil & structural design

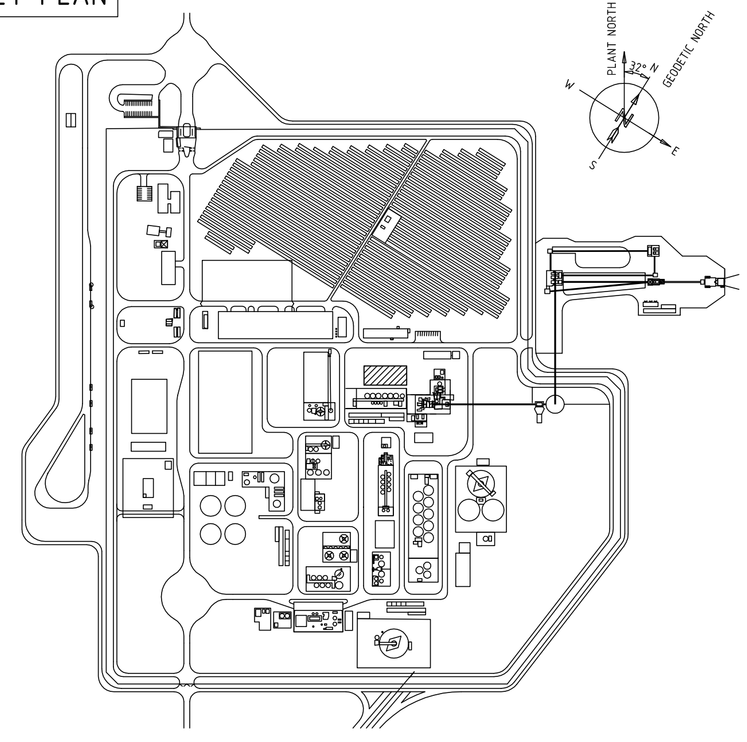
TABLE-3 : FOR DRIVE PULLEY FRAME LOAD

FOR CONV. 3110-CNV-001

POINTS	ELEVATIONS IN "m"	LOAD CASE I			LOAD CASE II		
		V (IN "t")	H ALONG (IN "t")	H ACROSS (IN "t")	V (IN "t")	H ALONG (IN "t")	H ACROSS (IN "t")
F3	(+)916.200	(+) 3.75	(-) 1.75	(±) 0.40	(+) 4.75	(-) 2.10	(±) 0.50
F4		(+) 2.70	(-) 2.10	(±) 0.30	(+) 1.75	(-) 2.00	(±) 0.20
F5		(+) 4.00	(-) 0.40	(±) 0.40	(-) 0.90	(-) 1.55	(±) 0.20

NOTE:
CASE I : CONVEYOR NORMAL RUNNING + MATERIAL LOAD + DEAD LOAD
CASE II : CONVEYOR STARTING + CHUTE EMPTY + DEAD LOAD

KEY PLAN



REV	DESCRIPTION	DRN	CHK	ENG	APP	EMP	DATE
D01	AS - BUILT	SM	BRD	BD	BGR		12/10/2023
C01	ISSUED FOR DETAIL DESIGN & REVISED AS PER EMPLOYER COMMENTS DT. 14.02.2020 & REVISED AS MARKED	SM	BRD	BD	BGR		18/03/2020
A01	ISSUED FOR APPROVAL	SM	BRD	BD	BGR		21/01/2020

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE
MD-216-3110-EG-ME-LPX-1006	TRANSFER POINT TP#1 - PLAN VIEW	A01	21/01/2020

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO

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Saudi Arabian Mining Company شركة التعدين العربية السعودية

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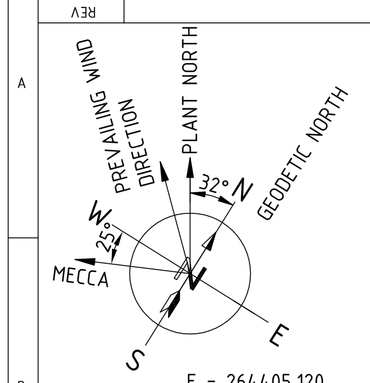
LARSEN & TOUBRO DRAWING NUMBER		REVISION
O19045_100_H_G_16130_002		2
OUTOTEC DRAWING NUMBER	DOCUMENT ID	REVISION
900707__GAU01__ZC01__MTL01__31148	OU100671062	2

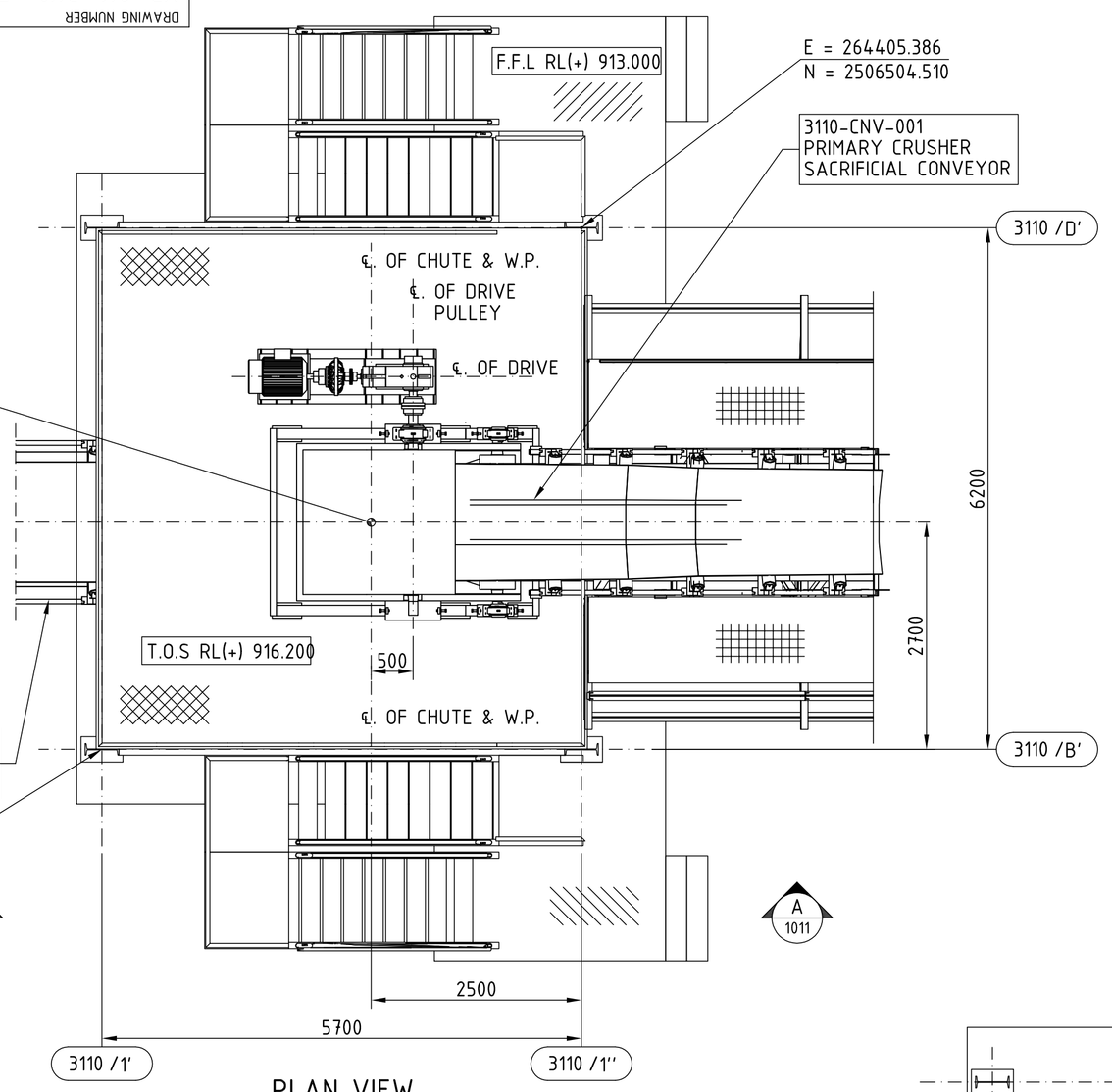
DRAWN	SM	18/01/2020
CHECKED	BRD	20/01/2020
ENGINEER	BD	20/01/2020
APPROVED	BGR	21/01/2020
EMPLOYER		
SCALE	1:50, 1:25 & 1;10	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
PRIMARY CRUSHING
TRANSFER POINT TP#1 - ELEVATION

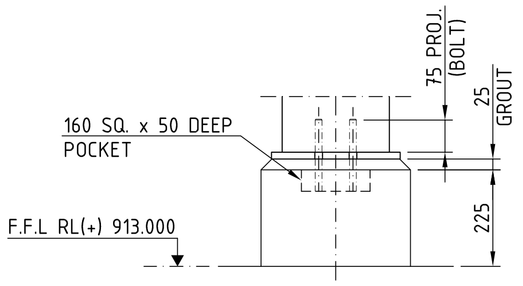
DRAWING NUMBER: MD-216-3110-EG-ME-ELV-1011 REV: D01

DRAWING NUMBER MD-216-3110-EG-ME-LPX-1006 REV D01

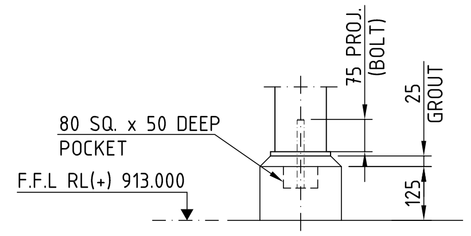




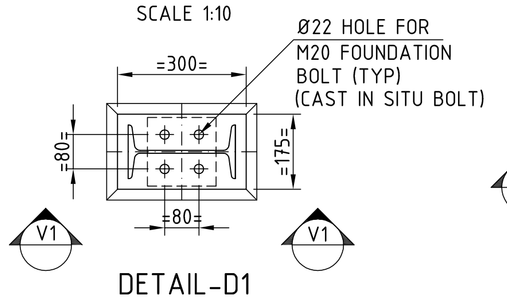
PLAN VIEW
SCALE 1:50



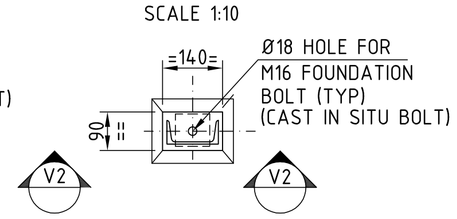
VIEW V1-V1
SCALE 1:10



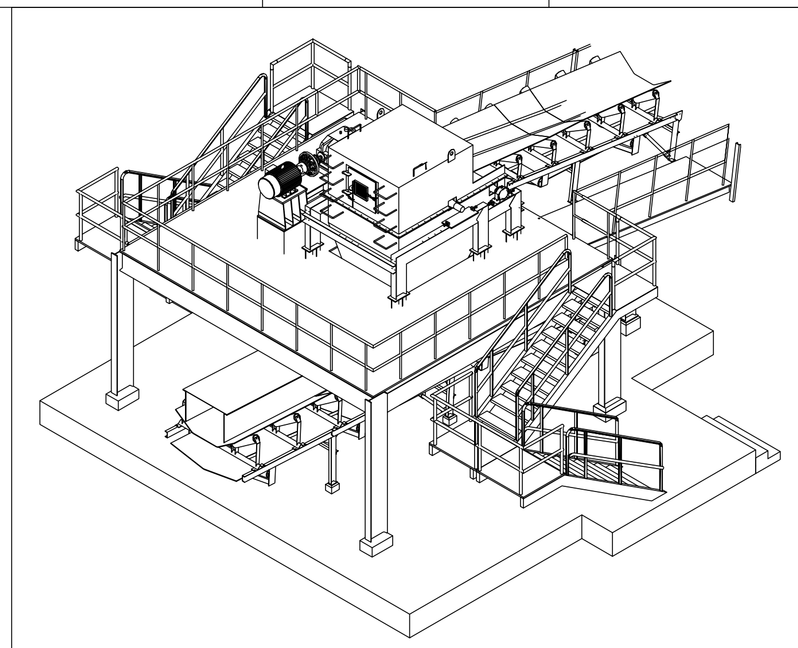
VIEW V2-V2
SCALE 1:10



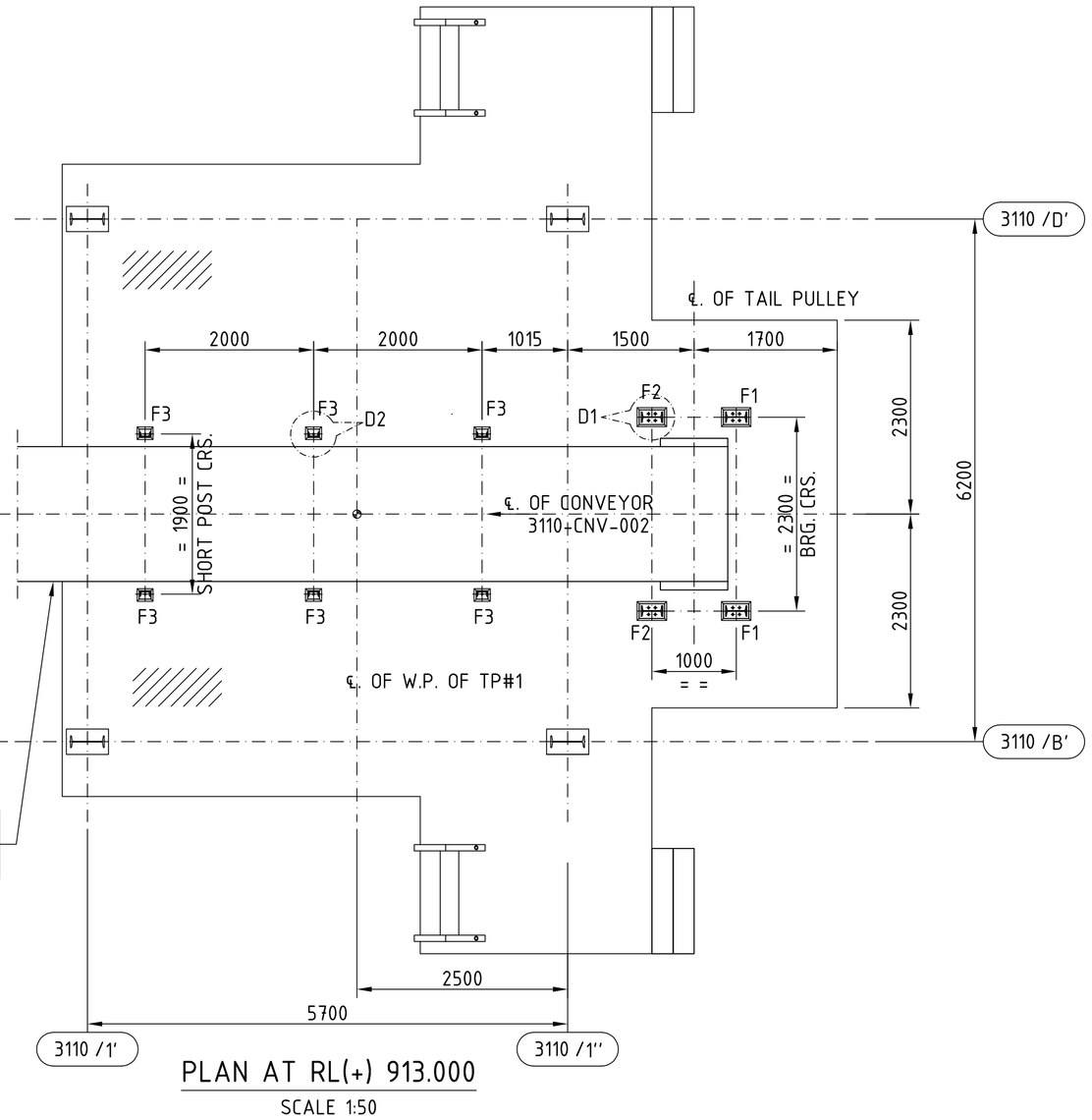
DETAIL-D1
SCALE 1:10



DETAIL-D2
SCALE 1:10



ISO VIEW

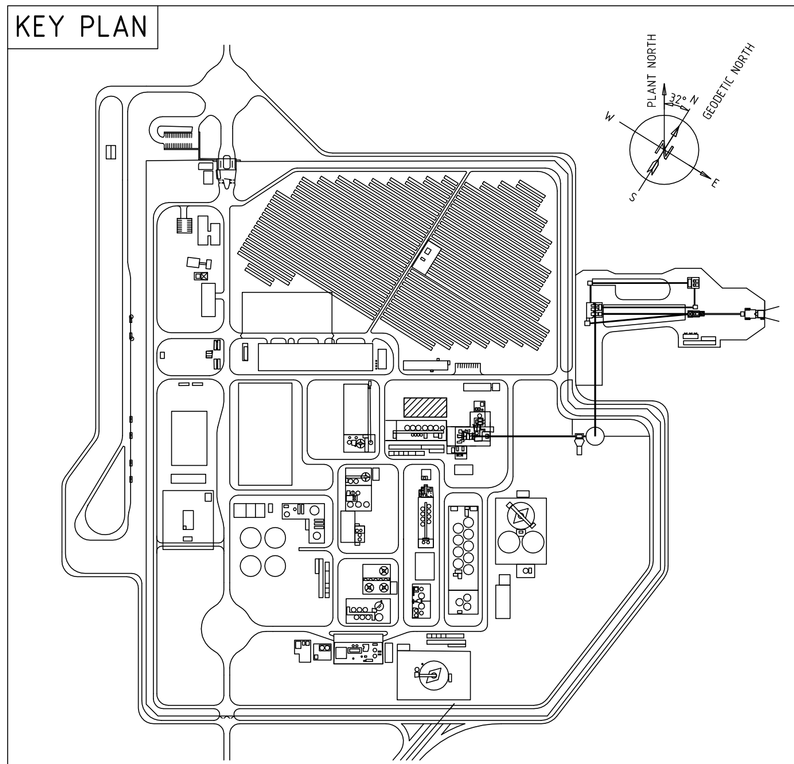


PLAN AT RL(+) 913.000
SCALE 1:50

NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+)913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL,BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN TENTATIVE.

CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

KEY PLAN



SIGN CONVENTION
(V) VERTICAL LOAD : UPWARD (-)
: DOWNWARD (+)
(H) HORIZONTAL LOAD : TOWARDS HEAD PULLEY=(+)
: TOWARDS TAIL PULLEY=(-)
: REVERSIBLE LOAD=(±)

Note: Wind load & seismic to be considered during civil & structural design

ABBREVIATION

CONV.	CONVEYOR
W.P.	WORKING POINT
℄	CENTER LINE
CRS.	CENTERS
B/W	BELT WIDTH
rpm	REVOLUTIONS PER MINUTE
BRG.	BEARING
SQ.	SQUARE
WT.	WEIGHT
PROJ.	PROJECTION
TP	TRANSFER POINT

TABLE-1 : FOR TAIL PULLEY FRAME LOAD
FOR CONV. 3110-CNV-002

POINTS	ELEVATIONS IN "m"	VERTICAL LOAD (IN "t")	HORIZONTAL LOAD (IN t)	
			ALONG THE CONV. (IN "t")	ACROSS THE CONV. (IN "t")
F1	(+)913.000	(-)4.00	(+)3.30	(±)0.2
F2		(+)6.00	(+)4.10	(±)0.2

TABLE-2 : LOAD FOR CONVEYOR SHORT POST
FOR CONV. 3110-CNV-002

	POINT	ELEVATION IN "m"	VERTICAL LOAD (IN "t")		HORIZONTAL LOAD (IN "t")	
			DEAD LOAD	LIVE LOAD	ALONG	ACROSS
IMPACT ZONE	F3	(+)913.000	(+) 1.10	(+) 1.00	(±) 0.21	(±) 0.11

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	12/10/2023	AS - BUILT	SM	BRD	BD	BGR	
		C01	18/03/2020	ISSUED FOR DETAIL DESIGN & REVISED AS PER EMPLOYER	SM	BRD	BD	BGR	
				COMMENTS DT. 14.02.2020 & REVISED AS MARKED					
MD-216-3110-EG-ME-ELV-1011	TRANSFER POINT TP#1 - ELEVATION	A01	21/01/2020	ISSUED FOR APPROVAL	SM	BRD	BD	BGR	

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LARSEN & TOUBRO DRAWING NUMBER		REVISION
019045_100_H_G_16130_001		2
OUTOTEC DRAWING NUMBER	DOCUMENT ID	REVISION
900707__GAU01__ZC01__MTL01__31147	OU100671061	2

DRAWN	SM	18/01/2020
CHECKED	BRD	20/01/2020
ENGINEER	BD	20/01/2020
APPROVED	BGR	21/01/2020
EMPLOYER		
SCALE	1:50, 1:10	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
PRIMARY CRUSHING
TRANSFER POINT TP#1 - PLAN VIEW

DRAWING NUMBER MD-216-3110-EG-ME-LPX-1006 REV D01

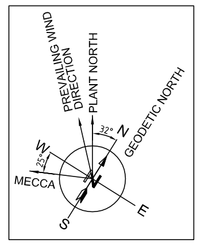
Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

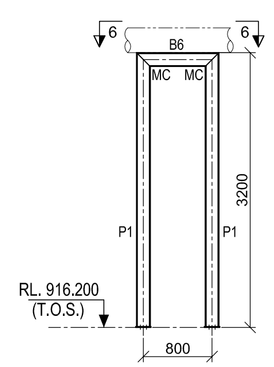
Ma'aden Revision No:	A01	C01	DO1
Date	10-04-2020	11-05-2020	12-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
Date	10-04-2020	11-05-2020	12-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	RPK	RPK	RPK
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

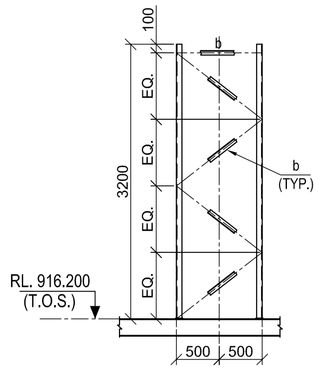
ATTACHMENT 2 **STRUCTURAL GENERAL ARRANGEMENT DRAWING CONTAINING BASE PLATE LAYOUT**

DRAWING NUMBER: MD-216-3120-EG-ST-GAD-1018
REV: A01

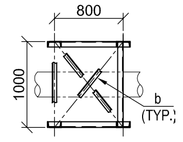




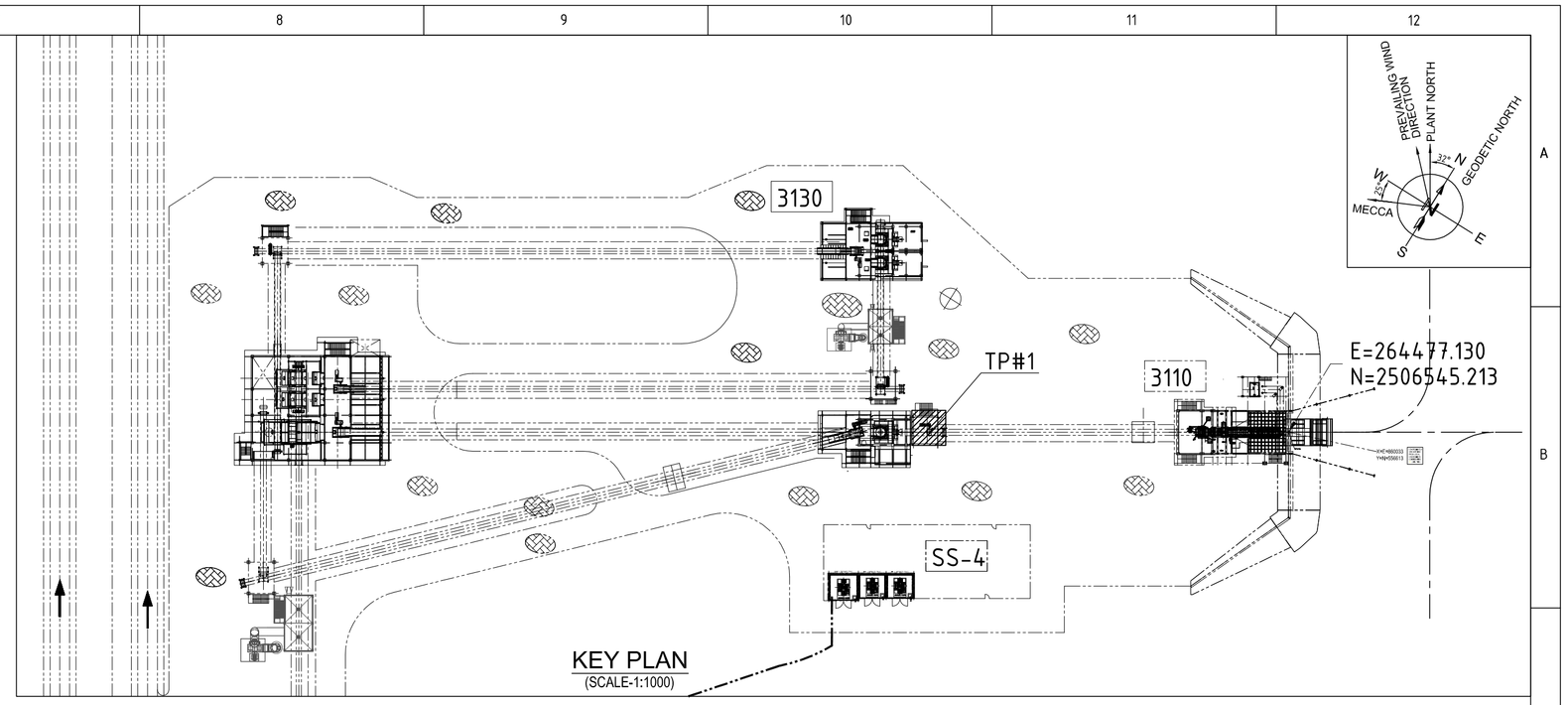
SECTION 5-5
(SCALE-1:50)



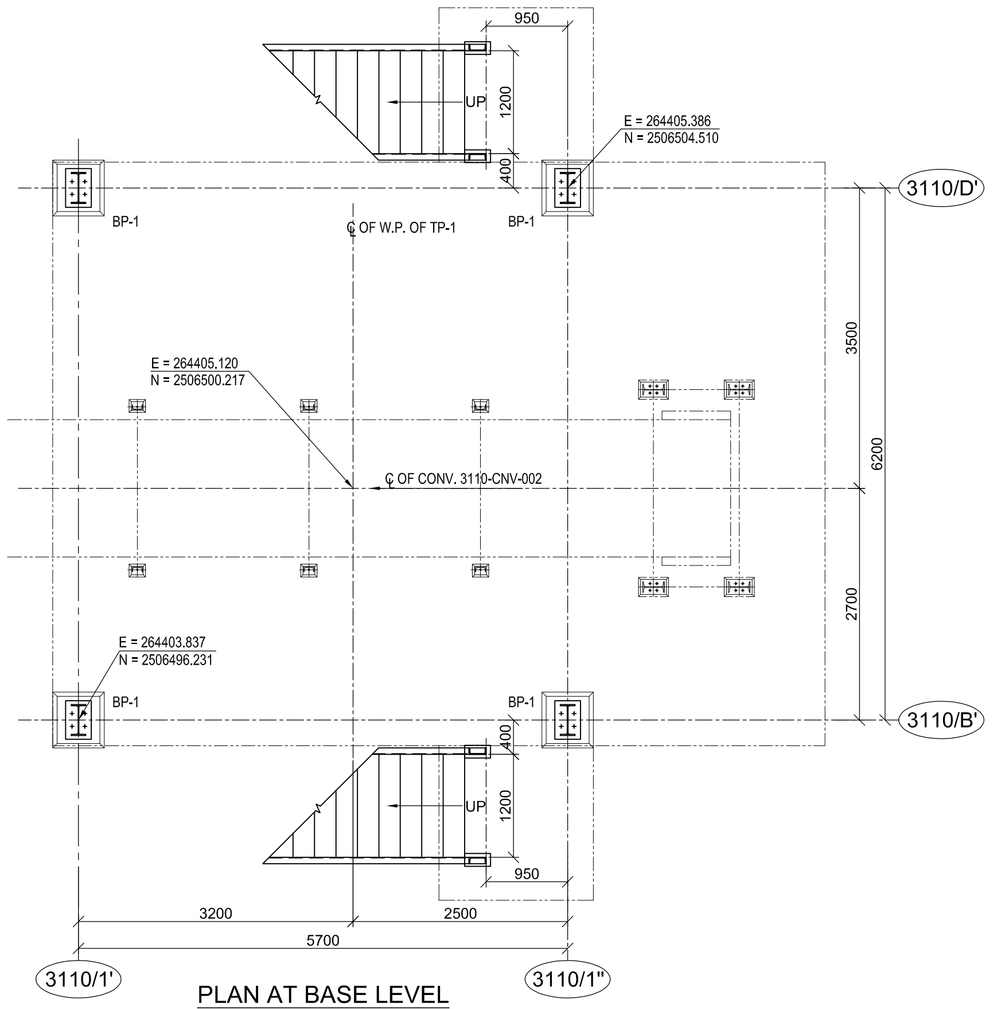
SECTION 5A-5A
(SCALE-1:50)



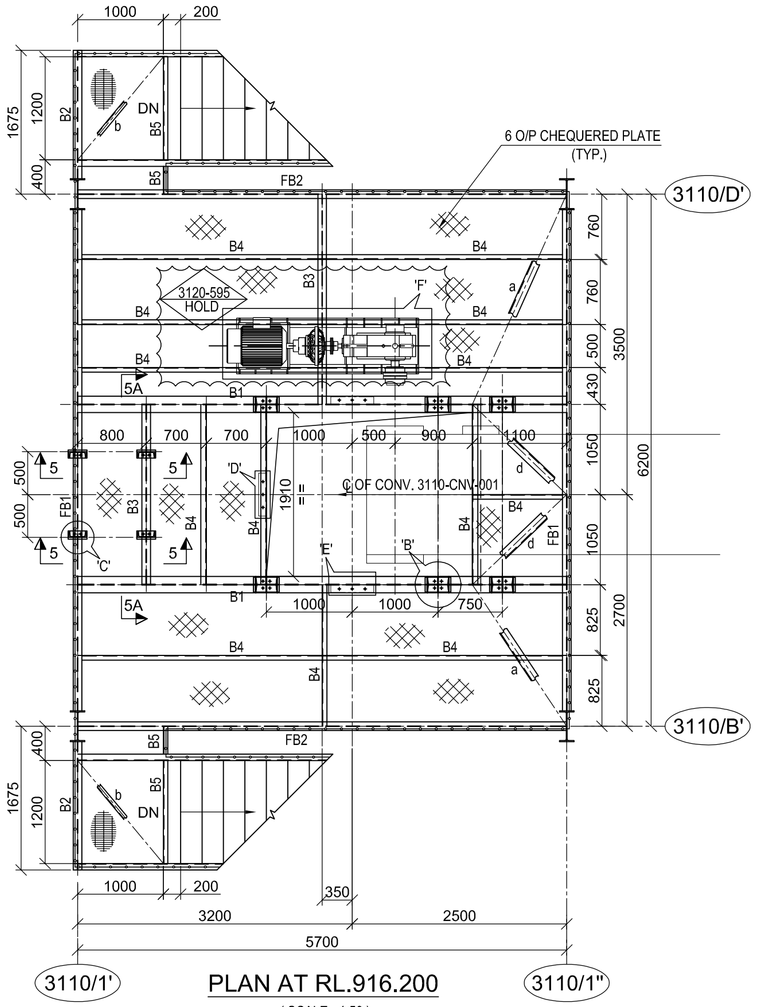
PLAN 6-6
(SCALE-1:50)



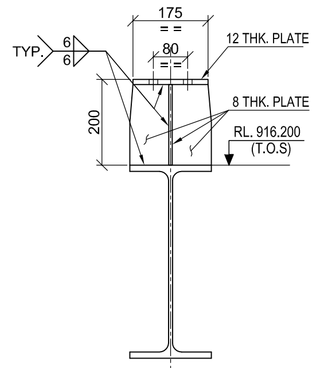
KEY PLAN
(SCALE-1:1000)



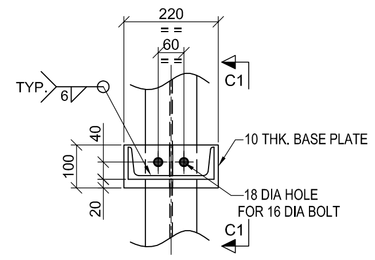
PLAN AT BASE LEVEL
(U/S OF BASE PLATE AT RL. 913.500 UNLESS NOTED OTHER WISE)
(SCALE - 1:50)



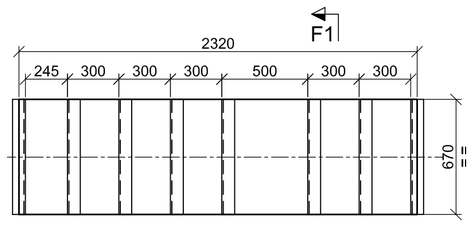
PLAN AT RL.916.200
(SCALE - 1:50)



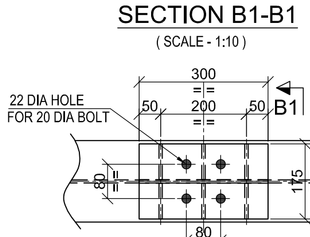
SECTION B1-B1
(SCALE - 1:10)



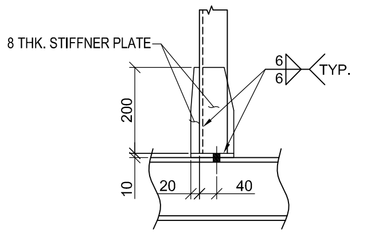
DETAIL 'C'
(REFER DRAWING NO. 1026)
(SCALE - 1:10)



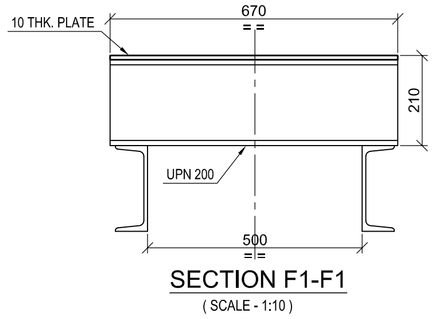
DETAIL 'F'
(SCALE - 1:25)



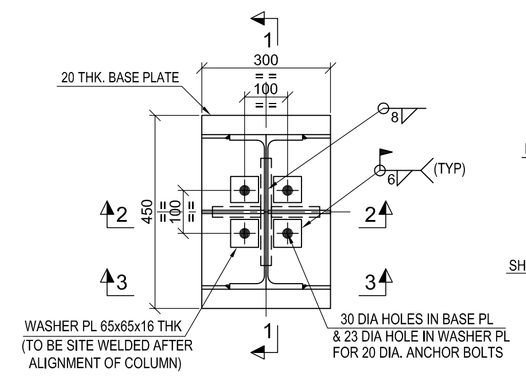
DETAIL 'B'
(SCALE - 1:10)



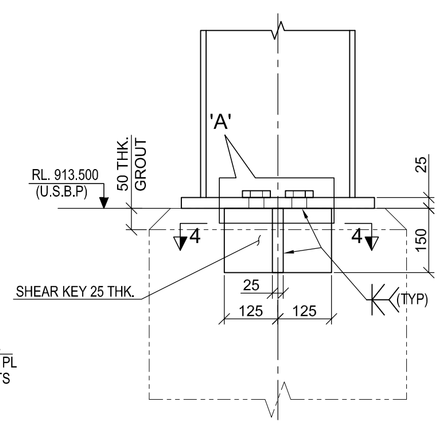
SECTION C1-C1
(SCALE - 1:10)



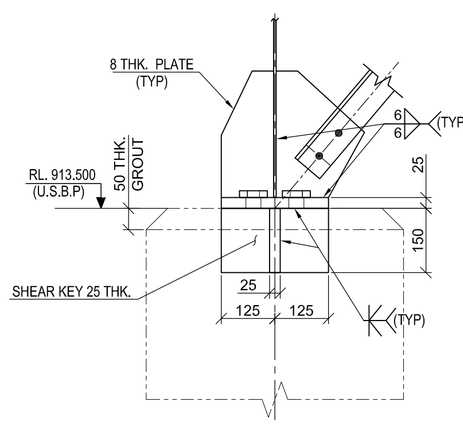
SECTION F1-F1
(SCALE - 1:10)



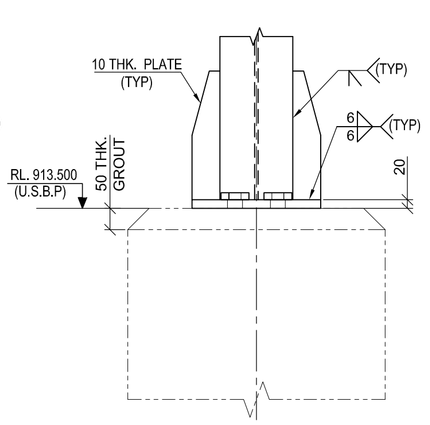
DETAIL OF BASE PLATE MKD. 'BP1'
(SCALE - 1:10)



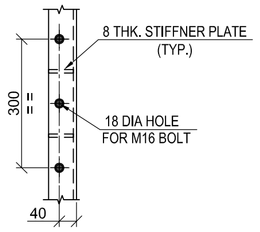
SECTION 1-1
(SCALE - 1:10)



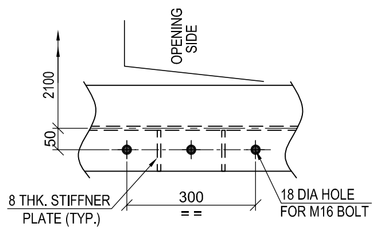
SECTION 2-2
(SCALE - 1:10)



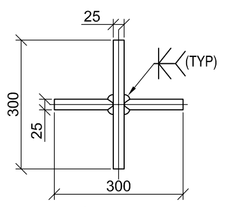
SECTION 3-3
(SCALE - 1:10)



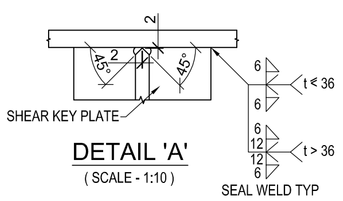
DETAIL 'D'
(SCALE - 1:10)



DETAIL 'E'
(SCALE - 1:10)



VIEW 4-4
(SCALE - 1:10)



DETAIL 'A'
(SCALE - 1:10)

SEAL WELD TYP

LEGEND :-
R.L. - REDUCED LEVEL
℄ - CENTRE LINE
T.O.S. - TOP OF STEEL.
F.G.L. - FINISHED GROUND LEVEL.
F.F.L. - FINISHED FLOOR LEVEL.
TYP. - TYPICAL.
MKD. - MARKED.
THK. - THICK
W.P.-WORKING POINT
U.S.B.P. - UNDER SIDE OF BASE PLATE
U.N.O. - UNLESS NOTED OTHERWISE

NOTES :-
1. ALL DIMENSIONS ARE IN MILLIMETER AND LEVELS & COORDINATES ARE IN METER UNO.
2. GRADE OF STEEL USED IS S275R AS PER EN10025.
3. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH DRAWING MD - 216 - 3120 -EG -ST - GAD - 1019.
4. ALL STRUCTURAL WORK SHALL CONFORM TO CIVIL STRUCTURAL DESIGN CRITARIYA NO MD - 216 - 1000 -EG -ST - CRT - 1001

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3110-EG-ME-LPX-1006	TRANSFER POINT TP#1 - PLAN VIEW								
MD-216-3110-EG-ME-ELEV-1011	TRANSFER POINT TP#1 - ELEVATION	A01	10/04/2020	ISSUED FOR REFERENCE	SUS	SBY	APR	PRI	

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LARSEN & TOUBRO DRAWING NUMBER: 019045_100_S_G_26204_001 — REVISION: 0
OUTOTEC DRAWING NUMBER: 900707__GAU01__ZC01__MTL01__31203 — DOCUMENT ID: 0U100668068 — REVISION: 0

DRAWN	SUS	08/04/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	SBY	09/04/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	APR	08/04/2020	STRUCTURAL GA OF TP-1(3120 CNV-001)-1
APPROVED	PRI	10/04/2020	PLAN, ELEVATIONS & DETAILS
EMPLOYER			
SCALE	1:100,20,15	A1	

DRAWING NUMBER: MD - 216 - 3120 - EG - ST - GAD - 1018 — REV: A01

MA'ADEN معادن شركة التعدين العربية السعودية Saudi Arabian Mining Company	Outotec	LARSEN & TOUBRO	**DESIGN DOCUMENT** **Ma'aden Document Number:** MD-216-3120-EG-CV-CAL-1002 Outotec Document Number: 900707_ GAU01_ZC01_CED01_31201_OU100668051 L&T Document Number: O19045_100_C_K_24111_001

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2		
Date	10-04-2020	11-05-2020	12-10-2023				Date	10-04-2020	11-05-2020	12-10-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	RPK	RPK	RPK				Prepared By	RPK	RPK	RPK		
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

ATTACHMENT 3 **STAAD INPUT & OUTPUT FILE OF TP-1 FOUNDATION**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 09-Apr-2020
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 5.7 1 6.2; 3 0 1 6.2; 5 5.7 1 12.4; 7 0 1 12.4; 14 5.7 -1.4 6.2;
15 0 -1.4 6.2; 16 5.7 -1.4 12.4; 17 0 -1.4 12.4; 18 5.7 0.5 6.2; 19 0 0.5 6.2;
20 0 0.5 12.4; 21 5.7 0.5 12.4;
MEMBER INCIDENCES
113 14 18; 114 15 19; 115 17 20; 116 16 21; 121 18 1; 122 19 3; 123 18 19;
124 20 7; 125 19 20; 126 21 5; 127 18 21; 128 21 20;
START GROUP DEFINITION
FLOOR
END GROUP DEFINITION
DEFINE MATERIAL START
ISOTROPIC STEEL
E 2.05e+008
POISSON 0.3
DENSITY 76.8195
ALPHA 1.2e-005
DAMP 0.03
TYPE STEEL
STRENGTH FY 253200 FU 407800 RY 1.5 RT 1.2
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
ISOTROPIC ALUMINUM
E 6.89476e+007
POISSON 0.33
DENSITY 26.6018
ALPHA 2.3e-005
DAMP 0.03
END DEFINE MATERIAL
MEMBER PROPERTY FRENCH
113 TO 116 121 122 124 126 PRIS YD 0.6 ZD 0.6
123 125 127 128 PRIS YD 0.5 ZD 0.3
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
14 TO 17 FIXED
LOAD 1 DEAD  LOAD
SELFWEIGHT Y -1
JOINT LOAD
1 FX 3 FY -24.8 FZ -3.6
3 FX -3.1 FY -18 FZ -3.7
5 FX 2.3 FY -26.2 FZ 3.6
7 FX -2.3 FY -18.5 FZ 3.7
```

```
LOAD 2 LIVE LOAD
JOINT LOAD
1 FX 9.8 FY -51.9 FZ -4.7
3 FX -9.9 FY -38.1 FZ -7.4
5 FX 6.9 FY -51.1 FZ 4.7
7 FX -6.9 FY -36.2 FZ 7.4
LOAD 3 SDL-1    EQUIPMENT DEAD LOAD CASE -1
JOINT LOAD
1 FX 25.4 FY -106.1 FZ -21.1
3 FX 8.1 FY -55.1 FZ -19.3
5 FX 25.9 FY -139 FZ 39.7
7 FX 25.6 FY -47.7 FZ 22.7
LOAD 4 SDL-2    EQUIPMENT DEAD LOAD CASE -2
JOINT LOAD
1 FX 34.1 FY -114.5 FZ -33.1
3 FX 17 FY -21.9 FZ -12.6
5 FX 31.1 FY -110.2 FZ 17.5
7 FX 30.9 FY -4.4 FZ 10.2
LOAD 5 WIND LOAD (+X)
JOINT LOAD
1 FX 6.53 FY -6.18 FZ -0.59
3 FX 6.54 FY 6.18 FZ 0.75
5 FX 7.48 FY -7.59 FZ 0.62
7 FX 7.48 FY 7.59 FZ -0.78
LOAD 6 WIND LOAD (-X)
JOINT LOAD
1 FX -6.53 FY 6.18 FZ 0.59
3 FX -6.54 FY -6.18 FZ -0.75
5 FX -7.48 FY 7.59 FZ -0.62
7 FX -7.48 FY -7.59 FZ 0.78
LOAD 7 WIND LOAD (+Z)
JOINT LOAD
1 FX -0.86 FY 7.19 FZ 7.05
3 FX -0.84 FY 5.65 FZ 6.39
5 FX 0.86 FY -7.09 FZ 6.74
7 FX 0.84 FY -5.75 FZ 6.01
LOAD 8 WIND LOAD (-Z)
JOINT LOAD
1 FX 0.86 FY -7.19 FZ -7.05
3 FX 0.84 FY -5.65 FZ -6.39
5 FX -0.86 FY 7.09 FZ -6.74
7 FX -0.84 FY 5.75 FZ -6.01
LOAD 9 TEMPERATURE LOAD 1
JOINT LOAD
1 FX 0.49 FY 0 FZ -1.73
3 FX -0.49 FY 0 FZ -1.73
5 FX 0.49 FY 0 FZ 1.73
7 FX -0.49 FY 0 FZ 1.73
LOAD 10 TEMPERATURE LOAD 2
JOINT LOAD
1 FX -0.49 FY 0 FZ 1.73
3 FX 0.49 FY 0 FZ 1.73
5 FX -0.49 FY 0 FZ -1.73
7 FX 0.49 FY 0 FZ -1.73
```

```
LOAD 11 SEISMIC LOAD (+X)
JOINT LOAD
1 FX 1.61 FY -1.54 FZ -0.02
3 FX 1.61 FY 1.54 FZ 0.01
5 FX 1.73 FY -1.67 FZ 0.03
7 FX 1.73 FY 1.67 FZ -0.02
LOAD 12 SEISMIC LOAD (+Z)
JOINT LOAD
1 FX -0.5 FY 2.6 FZ 2.46
3 FX -0.5 FY 0.35 FZ 0.89
5 FX 0.5 FY -2.6 FZ 2.46
7 FX 0.5 FY -0.35 FZ 0.88
***LOAD COMBINATION FOR BASE PRESSURE CHECK
LOAD 301 DL+SDL1
REPEAT LOAD
1 1.0 3 1.0
LOAD 302 DL+SDL2
REPEAT LOAD
1 1.0 4 1.0
LOAD 303 DL + SDL1 + TL1 + LL
REPEAT LOAD
1 1.0 3 1.0 9 1.0 2 1.0
LOAD 304 DL + SDL1 + TL1 + LL
REPEAT LOAD
1 1.0 4 1.0 9 1.0 2 1.0
LOAD 305 DL + SDL2 + TL2 + LL
REPEAT LOAD
1 1.0 3 1.0 10 1.0 2 1.0
LOAD 306 DL + SDL2 + TL2 + LL
REPEAT LOAD
1 1.0 4 1.0 10 1.0 2 1.0
LOAD 307 DL + SDL1 + 0.75LL + 0.75WL(+X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 5 0.75
LOAD 308 DL + SDL1 + 0.75LL + 0.75WL(-X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 6 0.75
LOAD 309 DL + SDL1 + 0.75LL + 0.75WL(+Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 7 0.75
LOAD 310 DL + SDL1 + 0.75LL + 0.75WL(-Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 8 0.75
LOAD 311 DL + SDL2 + 0.75LL + 0.75WL(+X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 5 0.75
LOAD 312 DL + SDL2 + 0.75LL + 0.75WL(-X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 6 0.75
LOAD 313 DL + SDL2 + 0.75LL + 0.75WL(+Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 7 0.75
LOAD 314 DL + SDL2 + 0.75LL + 0.75WL(-Z)
REPEAT LOAD
```

```
1 1.0 4 1.0 2 0.75 8 0.75
LOAD 315 DL + SDL1 + 0.75LL + 0.525EL(+X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 11 0.525
LOAD 316 DL + SDL1 + 0.75LL + 0.525EL(-X)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 11 -0.525
LOAD 317 DL + SDL1 + 0.75LL + 0.525EL(+Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 12 0.525
LOAD 318 DL + SDL1 + 0.75LL + 0.525EL(-Z)
REPEAT LOAD
1 1.0 3 1.0 2 0.75 12 -0.525
LOAD 319 DL + SDL2 + 0.75LL + 0.525EL(+X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 11 0.525
LOAD 320 DL + SDL2 + 0.75LL + 0.525EL(-X)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 11 -0.525
LOAD 321 DL + SDL2 + 0.75LL + 0.525EL(+Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 12 0.525
LOAD 322 DL + SDL2 + 0.75LL + 0.525EL(-Z)
REPEAT LOAD
1 1.0 4 1.0 2 0.75 12 -0.525
LOAD 323 0.6 (DL + SDL1) + WL(+X)
REPEAT LOAD
1 0.6 3 0.6 5 1.0
LOAD 324 0.6 (DL + SDL1) + WL(-X)
REPEAT LOAD
1 0.6 3 0.6 6 1.0
LOAD 325 0.6 (DL + SDL1) + WL(+Z)
REPEAT LOAD
1 0.6 3 0.6 7 1.0
LOAD 326 0.6 (DL + SDL1) + WL(-Z)
REPEAT LOAD
1 0.6 3 0.6 8 1.0
LOAD 327 0.6 (DL + SDL2) + WL(+X)
REPEAT LOAD
1 0.6 4 0.6 5 1.0
LOAD 328 0.6 (DL + SDL2) + WL(-X)
REPEAT LOAD
1 0.6 4 0.6 6 1.0
LOAD 329 0.6 (DL + SDL2) + WL(+Z)
REPEAT LOAD
1 0.6 4 0.6 7 1.0
LOAD 330 0.6 (DL + SDL2) + WL(-Z)
REPEAT LOAD
1 0.6 4 0.6 8 1.0
LOAD 331 0.6 (DL + SDL1) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 3 0.6 11 0.7
LOAD 332 0.6 (DL + SDL1) + 0.7 EL(-X)
REPEAT LOAD
```

```
1 0.6 3 0.6 11 -0.7
LOAD 333 0.6 (DL + SDL1) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 3 0.6 12 0.7
LOAD 334 0.6 (DL + SDL1) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 3 0.6 12 -0.7
LOAD 335 0.6 (DL + SDL2) + 0.7 EL(+X)
REPEAT LOAD
1 0.6 4 0.6 11 0.7
LOAD 336 0.6 (DL + SDL2) + 0.7 EL(-X)
REPEAT LOAD
1 0.6 4 0.6 11 -0.7
LOAD 337 0.6 (DL + SDL2) + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 4 0.6 12 0.7
LOAD 338 0.6 (DL + SDL2) + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 4 0.6 12 -0.7
****
*Load Combination for Concrete design
LOAD 401 1.4 (DL+SDL1)
REPEAT LOAD
1 1.4 3 1.4
LOAD 402 1.4 (DL+SDL2)
REPEAT LOAD
1 1.4 4 1.4
LOAD 403 1.4 (DL + SDL1 + TL1) + 1.7 LL
REPEAT LOAD
1 1.4 3 1.4 9 1.4 2 1.7
LOAD 404 1.4 (DL + SDL2 + TL1) + 1.7 LL
REPEAT LOAD
1 1.4 4 1.4 9 1.4 2 1.7
LOAD 405 1.4 (DL + SDL1 + TL2) + 1.7 LL
REPEAT LOAD
1 1.4 3 1.4 10 1.4 2 1.7
LOAD 406 1.4 (DL + SDL2 + TL2) + 1.7 LL
REPEAT LOAD
1 1.4 4 1.4 10 1.4 2 1.7
LOAD 407 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 5 1.6
LOAD 408 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 6 1.6
LOAD 409 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 7 1.6
LOAD 410 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 8 1.6
LOAD 411 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 5 1.6
```

```
LOAD 412 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 6 1.6
LOAD 413 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 7 1.6
LOAD 414 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 8 1.6
LOAD 415 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 11 1.0
LOAD 416 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 11 -1.0
LOAD 417 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 12 1.0
LOAD 418 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 3 1.2 2 0.5 12 -1.0
LOAD 419 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(+X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 11 1.0
LOAD 420 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 11 -1.0
LOAD 421 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(+Z)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 12 1.0
LOAD 422 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(-X)
REPEAT LOAD
1 1.2 4 1.2 2 0.5 12 -1.0
LOAD 423 0.9 (DL + SDL1) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 3 0.9 5 1.6
LOAD 424 0.9 (DL + SDL1) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 3 0.9 6 1.6
LOAD 425 0.9 (DL + SDL1) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 3 0.9 7 1.6
LOAD 426 0.9 (DL + SDL1) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 3 0.9 8 1.6
LOAD 427 0.9 (DL + SDL2) + 1.6 WL(+X)
REPEAT LOAD
1 0.9 4 0.9 5 1.6
LOAD 428 0.9 (DL + SDL2) + 1.6 WL(-X)
REPEAT LOAD
1 0.9 4 0.9 6 1.6
LOAD 429 0.9 (DL + SDL2) + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 0.9 7 1.6
```

```
LOAD 430 0.9 (DL + SDL2) + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 4 0.9 8 1.6
LOAD 431 0.9 (DL + SDL1) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 3 0.9 11 1.0
LOAD 432 0.9 (DL + SDL1) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 3 0.9 11 -1.0
LOAD 433 0.9 (DL + SDL1) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 3 0.9 12 1.0
LOAD 434 0.9 (DL + SDL1) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 3 0.9 12 -1.0
LOAD 435 0.9 (DL + SDL2) + 1.0 EL(+X)
REPEAT LOAD
1 0.9 4 0.9 11 1.0
LOAD 436 0.9 (DL + SDL2) + 1.0 EL(-X)
REPEAT LOAD
1 0.9 4 0.9 11 -1.0
LOAD 437 0.9 (DL + SDL2) + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 4 0.9 12 1.0
LOAD 438 0.9 (DL + SDL2) + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 4 0.9 12 -1.0
PERFORM ANALYSIS
LOAD LIST 401 TO 438
START CONCRETE DESIGN
CODE ACI 2008
FYMAIN 420000 ALL
FC 35000 ALL
MINMAIN 16 ALL
DESIGN COLUMN 113 TO 116 121 122 124 126
DESIGN BEAM 123 125 127 128
CLT 0.05 ALL
END CONCRETE DESIGN
FINISH
```

```
*****************************************************
*                                                   *
*           STAAD.Pro CONNECT Edition               *
*           Version  21.00.01.12                    *
*           Proprietary Program of                  *
*           Bentley Systems, Inc.                   *
*           Date=    APR 13, 2020                   *
*           Time=    10:24:29                       *
*                                                   *
*  Licensed to: MMH KOLKATTA                        *
*****************************************************
```

```
     1. STAAD SPACE
INPUT FILE: D:\F Drive\Maaden Gold\TP 1\Civil\Documents\Staad\TP-1 Civil.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 09-APR-2020
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 1 5.7 1 6.2; 3 0 1 6.2; 5 5.7 1 12.4; 7 0 1 12.4; 14 5.7 -1.4 6.2
     9. 15 0 -1.4 6.2; 16 5.7 -1.4 12.4; 17 0 -1.4 12.4; 18 5.7 0.5 6.2; 19 0 0.5 6.2
    10. 20 0 0.5 12.4; 21 5.7 0.5 12.4
    11. MEMBER INCIDENCES
    12. 113 14 18; 114 15 19; 115 17 20; 116 16 21; 121 18 1; 122 19 3; 123 18 19
    13. 124 20 7; 125 19 20; 126 21 5; 127 18 21; 128 21 20
    14. START GROUP DEFINITION
    15. FLOOR
    16. END GROUP DEFINITION

   ** WARNING: NO VALID GROUPS IN GROUP DEFINITION DATA **
    17. DEFINE MATERIAL START
    18. ISOTROPIC STEEL
    19. E 2.05E+008
    20. POISSON 0.3
    21. DENSITY 76.8195
    22. ALPHA 1.2E-005
    23. DAMP 0.03
    24. TYPE STEEL
    25. STRENGTH FY 253200 FU 407800 RY 1.5 RT 1.2
    26. ISOTROPIC CONCRETE
    27. E 2.17185E+007
    28. POISSON 0.17
    29. DENSITY 23.5616
    30. ALPHA 1E-005
    31. DAMP 0.05
    32. TYPE CONCRETE
    33. STRENGTH FCU 27579
    34. ISOTROPIC ALUMINUM
    35. E 6.89476E+007
    36. POISSON 0.33
```

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37. DENSITY 26.6018
38. ALPHA 2.3E-005
39. DAMP 0.03
40. END DEFINE MATERIAL
41. MEMBER PROPERTY FRENCH
42. 113 TO 116 121 122 124 126 PRIS YD 0.6 ZD 0.6
43. 123 125 127 128 PRIS YD 0.5 ZD 0.3
44. CONSTANTS
45. MATERIAL CONCRETE ALL
46. SUPPORTS
47. 14 TO 17 FIXED
48. LOAD 1 DEAD  LOAD
49. SELFWEIGHT Y -1
50. JOINT LOAD
51. 1 FX 3 FY -24.8 FZ -3.6
52. 3 FX -3.1 FY -18 FZ -3.7
53. 5 FX 2.3 FY -26.2 FZ 3.6
54. 7 FX -2.3 FY -18.5 FZ 3.7
55. LOAD 2 LIVE LOAD
56. JOINT LOAD
57. 1 FX 9.8 FY -51.9 FZ -4.7
58. 3 FX -9.9 FY -38.1 FZ -7.4
59. 5 FX 6.9 FY -51.1 FZ 4.7
60. 7 FX -6.9 FY -36.2 FZ 7.4
61. LOAD 3 SDL-1 EQUIPMENT DEAD LOAD CASE -1.
62. JOINT LOAD
63. 1 FX 25.4 FY -106.1 FZ -21.1
64. 3 FX 8.1 FY -55.1 FZ -19.3
65. 5 FX 25.9 FY -139 FZ 39.7
66. 7 FX 25.6 FY -47.7 FZ 22.7
67. LOAD 4 SDL-2 EQUIPMENT DEAD LOAD CASE -2
68. JOINT LOAD
69. 1 FX 34.1 FY -114.5 FZ -33.1
70. 3 FX 17 FY -21.9 FZ -12.6
71. 5 FX 31.1 FY -110.2 FZ 17.5
72. 7 FX 30.9 FY -4.4 FZ 10.2
73. LOAD 5 WIND LOAD (+X)
74. JOINT LOAD
75. 1 FX 6.53 FY -6.18 FZ -0.59
76. 3 FX 6.54 FY 6.18 FZ 0.75
77. 5 FX 7.48 FY -7.59 FZ 0.62
78. 7 FX 7.48 FY 7.59 FZ -0.78
79. LOAD 6 WIND LOAD (-X)
80. JOINT LOAD
81. 1 FX -6.53 FY 6.18 FZ 0.59
82. 3 FX -6.54 FY -6.18 FZ -0.75
83. 5 FX -7.48 FY 7.59 FZ -0.62
84. 7 FX -7.48 FY -7.59 FZ 0.78
85. LOAD 7 WIND LOAD (+Z)
86. JOINT LOAD
87. 1 FX -0.86 FY 7.19 FZ 7.05
88. 3 FX -0.84 FY 5.65 FZ 6.39
89. 5 FX 0.86 FY -7.09 FZ 6.74
90. 7 FX 0.84 FY -5.75 FZ 6.01
91. LOAD 8 WIND LOAD (-Z)
92. JOINT LOAD
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 93. 1 FX 0.86 FY -7.19 FZ -7.05
 94. 3 FX 0.84 FY -5.65 FZ -6.39
 95. 5 FX -0.86 FY 7.09 FZ -6.74
 96. 7 FX -0.84 FY 5.75 FZ -6.01
 97. LOAD 9 TEMPERATURE LOAD 1
 98. JOINT LOAD
 99. 1 FX 0.49 FY 0 FZ -1.73
100. 3 FX -0.49 FY 0 FZ -1.73
101. 5 FX 0.49 FY 0 FZ 1.73
102. 7 FX -0.49 FY 0 FZ 1.73
103. LOAD 10 TEMPERATURE LOAD 2
104. JOINT LOAD
105. 1 FX -0.49 FY 0 FZ 1.73
106. 3 FX 0.49 FY 0 FZ 1.73
107. 5 FX -0.49 FY 0 FZ -1.73
108. 7 FX 0.49 FY 0 FZ -1.73
109. LOAD 11 SEISMIC LOAD (+X)
110. JOINT LOAD
111. 1 FX 1.61 FY -1.54 FZ -0.02
112. 3 FX 1.61 FY 1.54 FZ 0.01
113. 5 FX 1.73 FY -1.67 FZ 0.03
114. 7 FX 1.73 FY 1.67 FZ -0.02
115. LOAD 12 SEISMIC LOAD (+Z)
116. JOINT LOAD
117. 1 FX -0.5 FY 2.6 FZ 2.46
118. 3 FX -0.5 FY 0.35 FZ 0.89
119. 5 FX 0.5 FY -2.6 FZ 2.46
120. 7 FX 0.5 FY -0.35 FZ 0.88
121. ***LOAD COMBINATION FOR BASE PRESSURE CHECK
122. LOAD 301 DL+SDL1
123. REPEAT LOAD
124. 1 1.0 3 1.0
125. LOAD 302 DL+SDL2
126. REPEAT LOAD
127. 1 1.0 4 1.0
128. LOAD 303 DL + SDL1 + TL1 + LL
129. REPEAT LOAD
130. 1 1.0 3 1.0 9 1.0 2 1.0
131. LOAD 304 DL + SDL1 + TL1 + LL
132. REPEAT LOAD
133. 1 1.0 4 1.0 9 1.0 2 1.0
134. LOAD 305 DL + SDL2 + TL2 + LL
135. REPEAT LOAD
136. 1 1.0 3 1.0 10 1.0 2 1.0
137. LOAD 306 DL + SDL2 + TL2 + LL
138. REPEAT LOAD
139. 1 1.0 4 1.0 10 1.0 2 1.0
140. LOAD 307 DL + SDL1 + 0.75LL + 0.75WL(+X)
141. REPEAT LOAD
142. 1 1.0 3 1.0 2 0.75 5 0.75
143. LOAD 308 DL + SDL1 + 0.75LL + 0.75WL(-X)
144. REPEAT LOAD
145. 1 1.0 3 1.0 2 0.75 6 0.75
146. LOAD 309 DL + SDL1 + 0.75LL + 0.75WL(+Z)
147. REPEAT LOAD
148. 1 1.0 3 1.0 2 0.75 7 0.75
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149. LOAD 310 DL + SDL1 + 0.75LL + 0.75WL(-Z)
150. REPEAT LOAD
151. 1 1.0 3 1.0 2 0.75 8 0.75
152. LOAD 311 DL + SDL2 + 0.75LL + 0.75WL(+X)
153. REPEAT LOAD
154. 1 1.0 4 1.0 2 0.75 5 0.75
155. LOAD 312 DL + SDL2 + 0.75LL + 0.75WL(-X)
156. REPEAT LOAD
157. 1 1.0 4 1.0 2 0.75 6 0.75
158. LOAD 313 DL + SDL2 + 0.75LL + 0.75WL(+Z)
159. REPEAT LOAD
160. 1 1.0 4 1.0 2 0.75 7 0.75
161. LOAD 314 DL + SDL2 + 0.75LL + 0.75WL(-Z)
162. REPEAT LOAD
163. 1 1.0 4 1.0 2 0.75 8 0.75
164. LOAD 315 DL + SDL1 + 0.75LL + 0.525EL(+X)
165. REPEAT LOAD
166. 1 1.0 3 1.0 2 0.75 11 0.525
167. LOAD 316 DL + SDL1 + 0.75LL + 0.525EL(-X)
168. REPEAT LOAD
169. 1 1.0 3 1.0 2 0.75 11 -0.525
170. LOAD 317 DL + SDL1 + 0.75LL + 0.525EL(+Z)
171. REPEAT LOAD
172. 1 1.0 3 1.0 2 0.75 12 0.525
173. LOAD 318 DL + SDL1 + 0.75LL + 0.525EL(-Z)
174. REPEAT LOAD
175. 1 1.0 3 1.0 2 0.75 12 -0.525
176. LOAD 319 DL + SDL2 + 0.75LL + 0.525EL(+X)
177. REPEAT LOAD
178. 1 1.0 4 1.0 2 0.75 11 0.525
179. LOAD 320 DL + SDL2 + 0.75LL + 0.525EL(-X)
180. REPEAT LOAD
181. 1 1.0 4 1.0 2 0.75 11 -0.525
182. LOAD 321 DL + SDL2 + 0.75LL + 0.525EL(+Z)
183. REPEAT LOAD
184. 1 1.0 4 1.0 2 0.75 12 0.525
185. LOAD 322 DL + SDL2 + 0.75LL + 0.525EL(-Z)
186. REPEAT LOAD
187. 1 1.0 4 1.0 2 0.75 12 -0.525
188. LOAD 323 0.6 (DL + SDL1) + WL(+X)
189. REPEAT LOAD
190. 1 0.6 3 0.6 5 1.0
191. LOAD 324 0.6 (DL + SDL1) + WL(-X)
192. REPEAT LOAD
193. 1 0.6 3 0.6 6 1.0
194. LOAD 325 0.6 (DL + SDL1) + WL(+Z)
195. REPEAT LOAD
196. 1 0.6 3 0.6 7 1.0
197. LOAD 326 0.6 (DL + SDL1) + WL(-Z)
198. REPEAT LOAD
199. 1 0.6 3 0.6 8 1.0
200. LOAD 327 0.6 (DL + SDL2) + WL(+X)
201. REPEAT LOAD
202. 1 0.6 4 0.6 5 1.0
203. LOAD 328 0.6 (DL + SDL2) + WL(-X)
204. REPEAT LOAD
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205. 1 0.6 4 0.6 6 1.0
206. LOAD 329 0.6 (DL + SDL2) + WL(+Z)
207. REPEAT LOAD
208. 1 0.6 4 0.6 7 1.0
209. LOAD 330 0.6 (DL + SDL2) + WL(-Z)
210. REPEAT LOAD
211. 1 0.6 4 0.6 8 1.0
212. LOAD 331 0.6 (DL + SDL1) + 0.7 EL(+X)
213. REPEAT LOAD
214. 1 0.6 3 0.6 11 0.7
215. LOAD 332 0.6 (DL + SDL1) + 0.7 EL(-X)
216. REPEAT LOAD
217. 1 0.6 3 0.6 11 -0.7
218. LOAD 333 0.6 (DL + SDL1) + 0.7 EL(+Z)
219. REPEAT LOAD
220. 1 0.6 3 0.6 12 0.7
221. LOAD 334 0.6 (DL + SDL1) + 0.7 EL(-Z)
222. REPEAT LOAD
223. 1 0.6 3 0.6 12 -0.7
224. LOAD 335 0.6 (DL + SDL2) + 0.7 EL(+X)
225. REPEAT LOAD
226. 1 0.6 4 0.6 11 0.7
227. LOAD 336 0.6 (DL + SDL2) + 0.7 EL(-X)
228. REPEAT LOAD
229. 1 0.6 4 0.6 11 -0.7
230. LOAD 337 0.6 (DL + SDL2) + 0.7 EL(+Z)
231. REPEAT LOAD
232. 1 0.6 4 0.6 12 0.7
233. LOAD 338 0.6 (DL + SDL2) + 0.7 EL(-Z)
234. REPEAT LOAD
235. 1 0.6 4 0.6 12 -0.7
236. ****
237. *LOAD COMBINATION FOR CONCRETE DESIGN
238. LOAD 401 1.4 (DL+SDL1)
239. REPEAT LOAD
240. 1 1.4 3 1.4
241. LOAD 402 1.4 (DL+SDL2)
242. REPEAT LOAD
243. 1 1.4 4 1.4
244. LOAD 403 1.4 (DL + SDL1 + TL1) + 1.7 LL
245. REPEAT LOAD
246. 1 1.4 3 1.4 9 1.4 2 1.7
247. LOAD 404 1.4 (DL + SDL2 + TL1) + 1.7 LL
248. REPEAT LOAD
249. 1 1.4 4 1.4 9 1.4 2 1.7
250. LOAD 405 1.4 (DL + SDL1 + TL2) + 1.7 LL
251. REPEAT LOAD
252. 1 1.4 3 1.4 10 1.4 2 1.7
253. LOAD 406 1.4 (DL + SDL2 + TL2) + 1.7 LL
254. REPEAT LOAD
255. 1 1.4 4 1.4 10 1.4 2 1.7
256. LOAD 407 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(+X)
257. REPEAT LOAD
258. 1 1.2 3 1.2 2 0.5 5 1.6
259. LOAD 408 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-X)
260. REPEAT LOAD
```

```
261. 1 1.2 3 1.2 2 0.5 6 1.6
262. LOAD 409 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(+Z)
263. REPEAT LOAD
264. 1 1.2 3 1.2 2 0.5 7 1.6
265. LOAD 410 1.2 (DL + SDL1) + 0.5 LL + 1.6 WL(-Z)
266. REPEAT LOAD
267. 1 1.2 3 1.2 2 0.5 8 1.6
268. LOAD 411 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+X)
269. REPEAT LOAD
270. 1 1.2 4 1.2 2 0.5 5 1.6
271. LOAD 412 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-X)
272. REPEAT LOAD
273. 1 1.2 4 1.2 2 0.5 6 1.6
274. LOAD 413 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(+Z)
275. REPEAT LOAD
276. 1 1.2 4 1.2 2 0.5 7 1.6
277. LOAD 414 1.2 (DL + SDL2) + 0.5 LL + 1.6 WL(-Z)
278. REPEAT LOAD
279. 1 1.2 4 1.2 2 0.5 8 1.6
280. LOAD 415 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(+X)
281. REPEAT LOAD
282. 1 1.2 3 1.2 2 0.5 11 1.0
283. LOAD 416 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(-X)
284. REPEAT LOAD
285. 1 1.2 3 1.2 2 0.5 11 -1.0
286. LOAD 417 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(+Z)
287. REPEAT LOAD
288. 1 1.2 3 1.2 2 0.5 12 1.0
289. LOAD 418 1.2 (DL + SDL1) + 0.5 LL + 1.0 EL(-X)
290. REPEAT LOAD
291. 1 1.2 3 1.2 2 0.5 12 -1.0
292. LOAD 419 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(+X)
293. REPEAT LOAD
294. 1 1.2 4 1.2 2 0.5 11 1.0
295. LOAD 420 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(-X)
296. REPEAT LOAD
297. 1 1.2 4 1.2 2 0.5 11 -1.0
298. LOAD 421 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(+Z)
299. REPEAT LOAD
300. 1 1.2 4 1.2 2 0.5 12 1.0
301. LOAD 422 1.2 (DL + SDL2) + 0.5 LL + 1.0 EL(-X)
302. REPEAT LOAD
303. 1 1.2 4 1.2 2 0.5 12 -1.0
304. LOAD 423 0.9 (DL + SDL1) + 1.6 WL(+X)
305. REPEAT LOAD
306. 1 0.9 3 0.9 5 1.6
307. LOAD 424 0.9 (DL + SDL1) + 1.6 WL(-X)
308. REPEAT LOAD
309. 1 0.9 3 0.9 6 1.6
310. LOAD 425 0.9 (DL + SDL1) + 1.6 WL(+Z)
311. REPEAT LOAD
312. 1 0.9 3 0.9 7 1.6
313. LOAD 426 0.9 (DL + SDL1) + 1.6 WL(-Z)
314. REPEAT LOAD
315. 1 0.9 3 0.9 8 1.6
316. LOAD 427 0.9 (DL + SDL2) + 1.6 WL(+X)
```

```
317. REPEAT LOAD
318. 1 0.9 4 0.9 5 1.6
319. LOAD 428 0.9 (DL + SDL2) + 1.6 WL(-X)
320. REPEAT LOAD
321. 1 0.9 4 0.9 6 1.6
322. LOAD 429 0.9 (DL + SDL2) + 1.6 WL(+Z)
323. REPEAT LOAD
324. 1 0.9 4 0.9 7 1.6
325. LOAD 430 0.9 (DL + SDL2) + 1.6 WL(-Z)
326. REPEAT LOAD
327. 1 0.9 4 0.9 8 1.6
328. LOAD 431 0.9 (DL + SDL1) + 1.0 EL(+X)
329. REPEAT LOAD
330. 1 0.9 3 0.9 11 1.0
331. LOAD 432 0.9 (DL + SDL1) + 1.0 EL(-X)
332. REPEAT LOAD
333. 1 0.9 3 0.9 11 -1.0
334. LOAD 433 0.9 (DL + SDL1) + 1.0 EL(+Z)
335. REPEAT LOAD
336. 1 0.9 3 0.9 12 1.0
337. LOAD 434 0.9 (DL + SDL1) + 1.0 EL(-Z)
338. REPEAT LOAD
339. 1 0.9 3 0.9 12 -1.0
340. LOAD 435 0.9 (DL + SDL2) + 1.0 EL(+X)
341. REPEAT LOAD
342. 1 0.9 4 0.9 11 1.0
343. LOAD 436 0.9 (DL + SDL2) + 1.0 EL(-X)
344. REPEAT LOAD
345. 1 0.9 4 0.9 11 -1.0
346. LOAD 437 0.9 (DL + SDL2) + 1.0 EL(+Z)
347. REPEAT LOAD
348. 1 0.9 4 0.9 12 1.0
349. LOAD 438 0.9 (DL + SDL2) + 1.0 EL(-Z)
350. REPEAT LOAD
351. 1 0.9 4 0.9 12 -1.0
352. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**12**	**NUMBER OF MEMBERS**	**12**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**4**

Using 64-bit analysis engine.

SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL PRIMARY LOAD CASES = 88, TOTAL DEGREES OF FREEDOM = 48
TOTAL LOAD COMBINATION CASES = 0 SO FAR.

```
353. LOAD LIST 401 TO 438
354. START CONCRETE DESIGN
355. CODE ACI 2008
356. FYMAIN 420000 ALL
357. FC 35000 ALL
358. MINMAIN 16 ALL
359. DESIGN COLUMN 113 TO 116 121 122 124 126
```

==========

COLUMN NO. 113 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 114 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 115 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 116 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 121 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 122 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 124 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

==========

COLUMN NO. 126 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 320.00 MM

**********END OF COLUMN DESIGN RESULTS**********

360. DESIGN BEAM 123 125 127 128

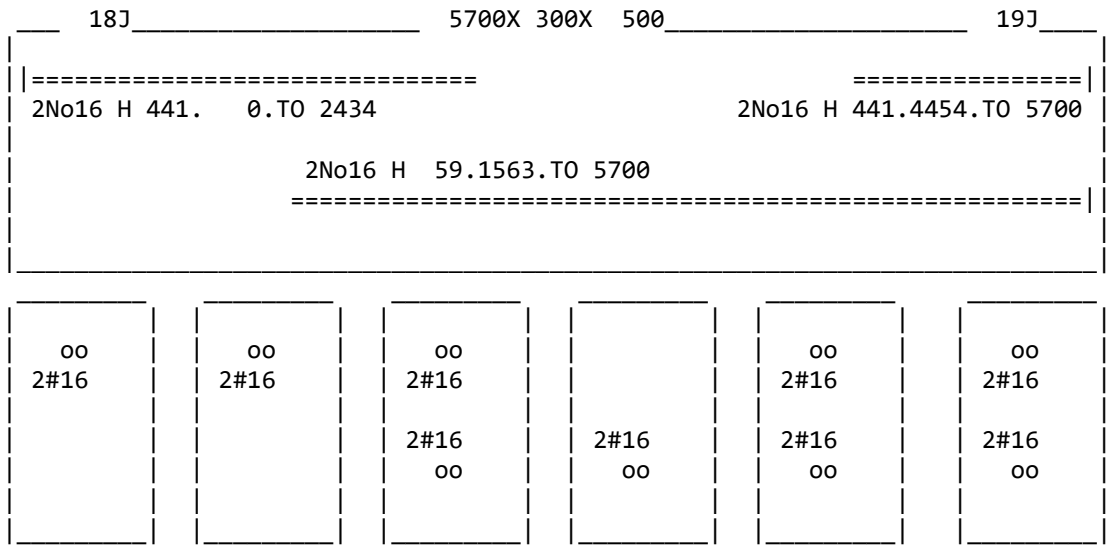
BEAM NO. 123 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5700. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	59.	2 - 16MM	1563.	5700.	NO	YES
2	441.	2 - 16MM	0.	2434.	YES	NO
3	441.	2 - 16MM	4454.	5700.	NO	YES

B E A M N O. 123 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 16.25 KNS Vc= 127.33 KNS Vs= 0.00 KNS
Tu= 0.24 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 7.63 KNS Vc= 127.33 KNS Vs= 0.00 KNS
Tu= 0.24 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



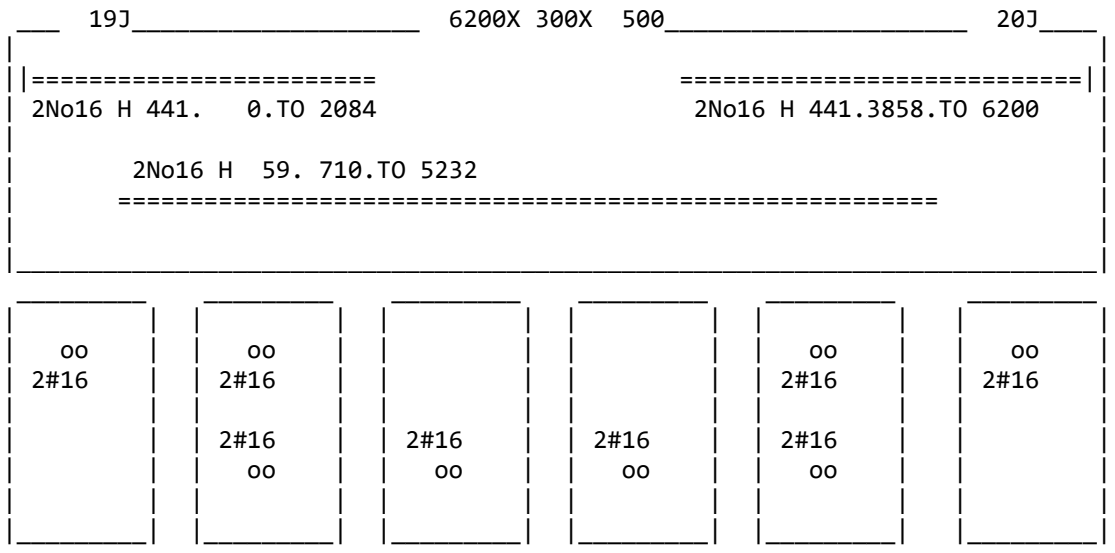
BEAM NO. 125 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 6200. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	59.	2 - 16MM	710.	5232.	NO	NO
2	441.	2 - 16MM	0.	2084.	YES	NO
3	441.	2 - 16MM	3858.	6200.	NO	YES

B E A M N O. 125 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 12.73 KNS Vc= 122.09 KNS Vs= 0.00 KNS
Tu= 0.48 KN-MET Tc= 4.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 13.62 KNS Vc= 122.09 KNS Vs= 0.00 KNS
Tu= 0.48 KN-MET Tc= 4.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



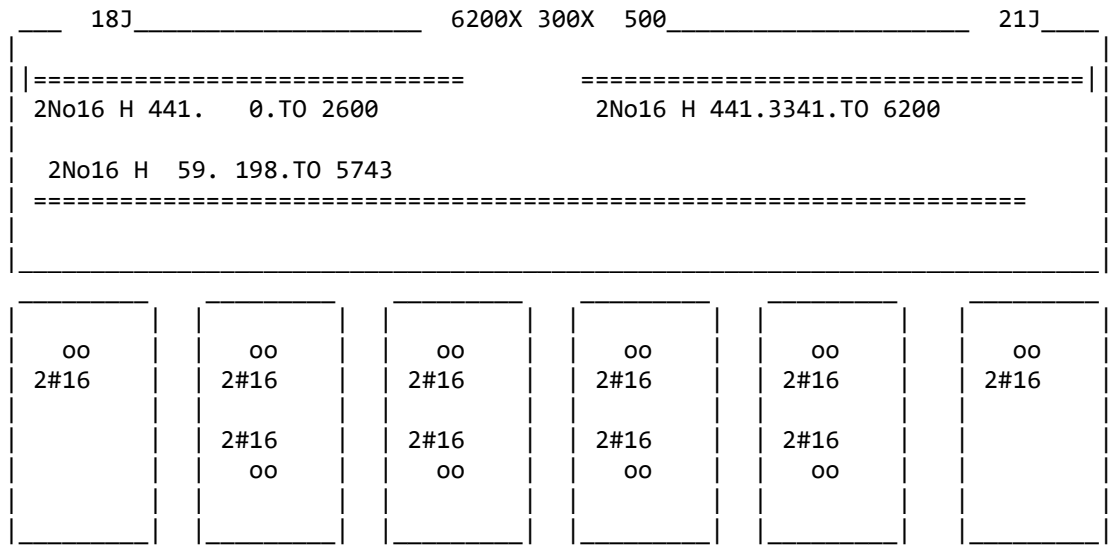
BEAM NO. 127 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 6200. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	59.	2 - 16MM	198.	5743.	NO	NO
2	441.	2 - 16MM	0.	2600.	YES	NO
3	441.	2 - 16MM	3341.	6200.	NO	YES

B E A M N O. 127 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 11.17 KNS Vc= 118.12 KNS Vs= 0.00 KNS
Tu= 0.28 KN-MET Tc= 4.7 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 15.18 KNS Vc= 118.12 KNS Vs= 0.00 KNS
Tu= 0.28 KN-MET Tc= 4.7 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



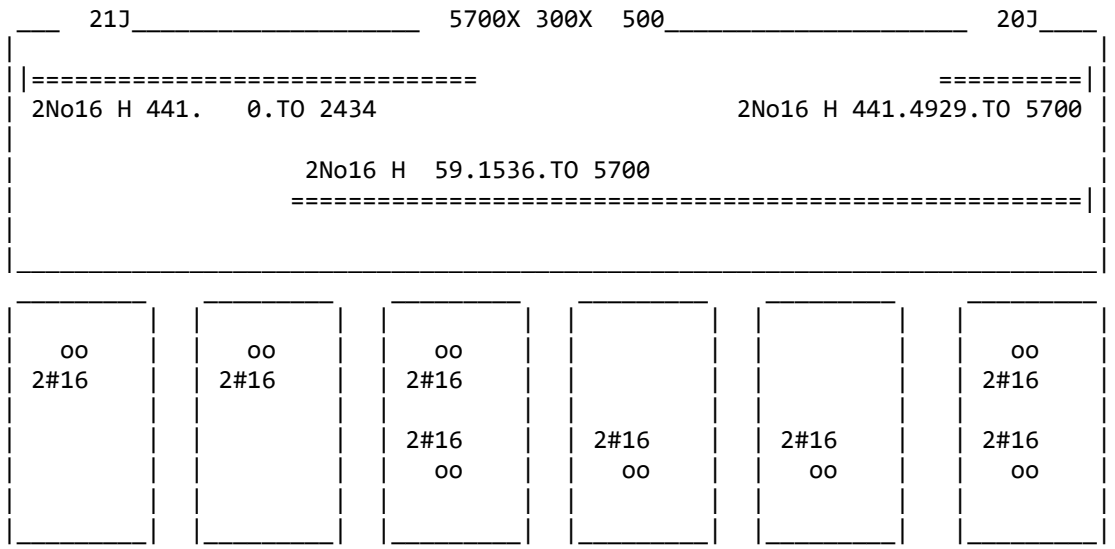
BEAM NO. 128 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5700. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	59.	2 - 16MM	1536.	5700.	NO	YES
2	441.	2 - 16MM	0.	2434.	YES	NO
3	441.	2 - 16MM	4929.	5700.	NO	YES

B E A M N O. 128 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 18.41 KNS Vc= 124.53 KNS Vs= 0.00 KNS
Tu= 0.47 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 5.48 KNS Vc= 124.40 KNS Vs= 0.00 KNS
Tu= 0.47 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



**********END OF BEAM DESIGN**********

361. CLT 0.05 ALL

```
362. END CONCRETE DESIGN
363. FINISH


     ********** END OF THE STAAD.Pro RUN **********

       **** DATE= APR 13,2020   TIME= 10:24:34 ****

   ************************************************************
   *   For technical assistance on STAAD.Pro, please visit    *
   *   http://selectservices.bentley.com/en-US/               *
   *                                                          *
   *   Details about additional assistance from               *
   *   Bentley and Partners can be found at program menu      *
   *   Help->Technical Support                                *
   *                                                          *
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